Hubbard dynamics on a fixed qubit array: protection, mitigation, and achievable lattice size

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Abstract

Error mitigation, partial protection, and full fault tolerance compete for the same physical qubits, and the comparison between them is decided by choices that are usually fixed before it begins. In this work, we compare the three on one nearest-neighbor array for a local doublon observable after a Fermi–Hubbard quench, with one week of execution and an evolution time that grows with the lattice. A common accounting model separates sampling, mitigation exposure, logical protection, non-Clifford supply, and parallel replicas, and optimizes each architecture's step count, error allocation, and delivery. We then promote five previously fixed choices to optimization variables: raw repeat-until-success injection, analog parallelism, fault-tolerant T delivery, separate source and data distances, and Richardson extrapolation of the Trotter error. Joint optimization of T-consumption ports, factories, and Hamming-weight phasing changes the leader most often. With all five choices at a physical error of 10^{-3}, full fault tolerance leads beyond about 6×10^5 physical qubits; at an absolute error of 10^{-3}, this point moves from 1.6×10^7 to 6×10^5 qubits. STAR, and a hybrid that draws only its fixed-angle rotations from magic-state cultivation, saturate near $\tau = 2$–3, and no STAR-only window remains. These orderings are conditional on declared delivery models, finite-patch Trotter transfer, a reference surface-code law shared by both encoded arms, and an assumed mitigation overhead for physical qubits; a circuit-level calibration of that law raises the encoded patch area by a factor of about 1.4 at $p = 10^{-3}$ and is not yet propagated. Published resource ledgers and experimental data check the accounting on their original tasks. Physical qubits pay per gate, STAR per radian, and full fault tolerance per logarithm; for this workload, the radians grow faster.

1 Which choices change the leader

The integrated comparison of Sec. 2 fixed several choices that control its ordering: four analog lanes, one T-consumption port, Hamming-weight batches of at most 16 rotations, a data distance no smaller than the cultivation output distance, noisy probabilistic coherent error cancellation (PCEC) for every analog rotation, and a single Trotter branch. It also measured capacity as the largest fully evolved lattice at fixed $\tau = 1$. In this section, we replace that headline by a time-growing workload, promote the five choices to optimization variables, and ask which of them changes the capacity leader. Each change is tested alone and all five together against a reference menu that reproduces the integrated model; 335 of 336 replayed cases agree to 10^{-9} in $\log_{10} Q$. The remaining case, a STAR design costed at $10^{78.7}$ qubits, is removed by an exposure cutoff ($\chi > 120$) that cannot affect capacity below 10^9 qubits.

The headline workload grows with the lattice. A local observable at fixed τ stops depending on L once the lattice contains its light cone. For free fermions ($U = 0$), the central doublon at $\tau = 1$ differs from its $L = 60$ value by 2.4×10^{-3} at $L = 6$ and by 2.4×10^{-4} at $L = 8$. Fixed-time capacity beyond

$L \approx 8$ therefore measures the cost of evolving a full lattice, which remains a useful implementation benchmark, rather than a more converged answer. We lead with $\tau = L/4$ for $L = 4, 8, \ldots, 32$, for which the relevant region and the lattice grow together. At $\tau \in \{1, 2, 4\}$ the Trotter coefficient is the measured 4×4 envelope used in Sec. 2; at other times we apply the same running-maximum rule to every measured patch through τ. We stress that the free-fermion numbers motivate this choice but do not bound interacting dynamics; interacting bulk convergence and patches larger than 4×4 remain open.

Change	Optimized quantity	Declared assumption
Raw RUS	Raw repeat-until-success (RUS) channel beside noisy PCEC for STAR and hybrid rotations	Same generator-fault instrument; coherent residue and retry tail charged to the error budget
Lanes	Analog lanes, 1–128	Six tiles per lane; at most L lanes for termwise Jordan–Wigner rotations and N for plaquette rotations; more than four lanes use the fast block
FT delivery	T-consumption ports (1–128), factories, and Hamming-weight batches up to 1024, jointly	Two tiles per extra port; ports use the fast block; serial rotation sequences, Eq. (2); no analog-lane tiles in the FT block
Distances	Data distance separate from the cultivation output distance	Transfer charged as $2\max(d, 15)$ cycles per state; STAR receivers keep their measured growth targets
Richardson	Single branch or branches $(r, 2r)$ with coefficients $(-\frac{1}{3}, \frac{4}{3})$	Branch-specific time, variance, mitigation, and bias, Eq. (1)

Table 1: The five changes tested in this section. Each enters the same optimizer over step count, error allocation, and source menu used in Sec. 2. Tile charges and concurrency limits are declared delivery models, not compiled schedules.

Richardson branches are costed individually. Consider two product-formula branches with step counts r and $2r$ and coefficients $c = (-\frac{1}{3}, \frac{4}{3})$, which cancel the leading r^{-2} error of the second-order formula [1, 2]. With per-attempt variance multipliers $F_i = e^{\chi_i}$ and attempt times t_i, the optimal allocation of shots between branches gives a total sampling time

$$T_{\text{sample}} = \frac{N_0}{m}\Big(\sum_i |c_i|\sqrt{F_i\, t_i}\Big)^2, \tag{1}$$

where N_0 is the ideal shot count and m the number of replicas. With equal F and $t_2 = 2t_1$, Eq. (1) costs $(\frac{1}{3} + \frac{4}{3}\sqrt{2})^2 = 4.92$ base-branch runtimes, so extrapolation pays only when the Trotter allowance it frees buys more than that. The branches share hardware and code distance. Branch biases add with weights $|c_i|$, so the logical, magic, and synthesis allowances are divided by $\|c\|_1 = 5/3$ after each branch's own Γ amplification. The residual Trotter error is K_4/r^4. On the 4×4 plaquette patch at $\tau = 1$, the measured $|4E(2r) - E(r)|r^4/3$ equals 0.0514 at both $r = 8$ and $r = 16$, so the residual is cleanly fourth order, as expected for a symmetric formula [3]. We use 1.1 times the running maximum of this quantity over all signed same-patch pairs through τ, and keep the base step count at $r \geq 8\tau$. For physical qubits, each branch pays its own mitigation exponent.

Ports cannot parallelize a synthesized rotation. A synthesized rotation is a serial T sequence, whereas the rotations of one plaquette stage are independent. With P consumption ports, the serial

T depth of one shot is

$$N_T^{\text{serial}} = n_T \sum_s m_s \left\lceil \frac{R_s}{P} \right\rceil + \frac{N_T - n_T R}{P}, \tag{2}$$

where s labels plaquette stages, each repeated m_s times with R_s synthesized rotations, n_T is the T count per rotation, $R = \sum_s m_s R_s$, and N_T is the total T count including fixed-angle and Toffoli states. At $P = 1$, Eq. (2) reduces to N_T and reproduces the integrated model. Larger Hamming-weight batches therefore trade parallel width for fewer synthesized rotations, and the optimizer must buy ports, factories, and batch size together. Additional factories are not cheap: with one cleanup level, each costs $1024 + 13 \cdot 2d^2$ qubits, or 22,890 at $d = 29$.

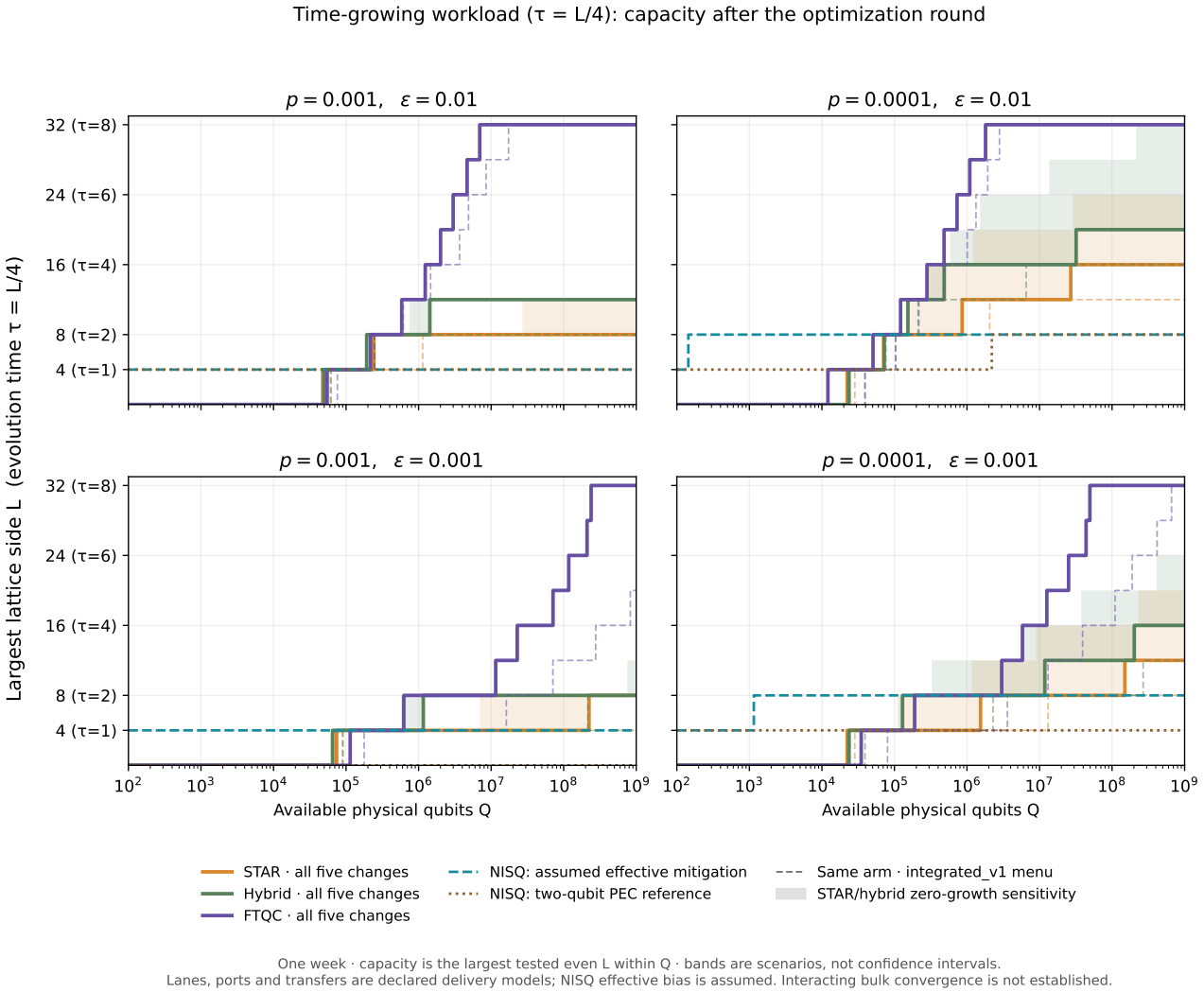


Figure 1: Capacity for the time-growing workload $\tau = L/4$ after the optimization round. Solid curves use all five changes; thin dashed curves of the same color use the reference menu. Bands give the STAR and hybrid zero-growth sensitivity. NISQ curves assume the effective mitigation overhead of Sec. 7 or apply two-qubit PEC. $L = 32$ ($\tau = 8$) is the sampled ceiling. Delivery models are declared, and interacting bulk convergence is not established.

Full fault tolerance separates earlier. Figure 1 shows the time-growing capacity. At $p = 10^{-3}$ and $\epsilon = 0.01$, encoded computation overtakes the effective NISQ line at about 2×10^5 physical qubits, and full fault tolerance leads alone beyond 6×10^5 qubits, reaching the $\tau = 8$ ceiling at 7×10^6. STAR saturates at $L = 8$ ($\tau = 2$) and the hybrid at $L = 12$ ($\tau = 3$). Physical-qubit computation stays at $\tau = 1$ even with the assumed effective overhead. At $\epsilon = 0.001$, the point beyond which full fault tolerance leads moves from 1.6×10^7 qubits under the reference menu to 6×10^5. At $p = 10^{-4}$ the same point moves from 1.3×10^7 to 3×10^6 qubits at $\epsilon = 0.001$, and the analog arms saturate later, at $L = 16$ (STAR) and $L = 20$ (hybrid) for $\epsilon = 0.01$. The hybrid injects the variable plaquette angles in analog and draws its fixed $\pi/8$ rotations from the same cultivated source as full fault tolerance; with all five changes it never falls below STAR.

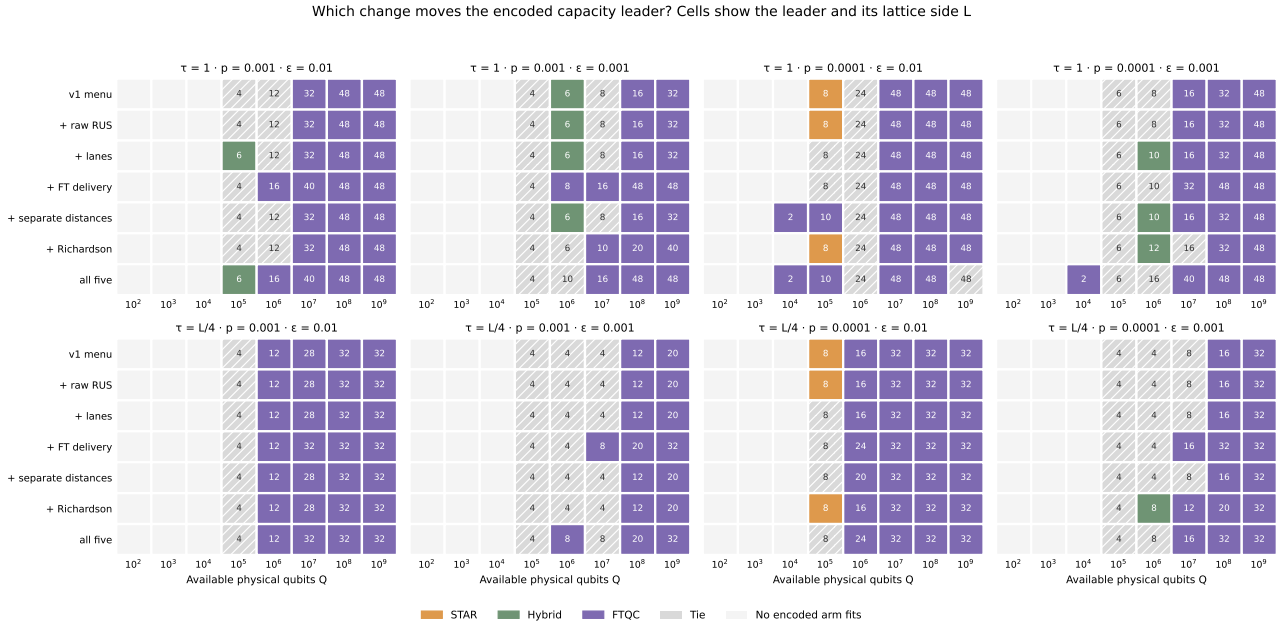


Figure 2: Encoded capacity leader for each configuration (rows), qubit budget (columns), time family, physical error, and precision. Numbers give the leader's lattice side; hatched cells are ties. The reference menu is the top row of each panel.

Delivery, not injection, moves the leader most often. Figure 2 and Table 2 attribute the changes. Out of 64 cells, joint T delivery changes the leader in 8. At 48×48, $\tau = 1$, and $\epsilon = 0.001$, whole-group phasing with batches of 512 reduces the T count from 1.34×10^7 to 3.1×10^6, while 32 ports and 1024 factories reduce the attempt time from 105 s to 2 s; the required area falls from $10^{9.4}$ to $10^{7.9}$ qubits although factories occupy 57% of it. Richardson extrapolation changes six cells, all at $\epsilon = 0.001$, and helps every encoded arm. Separate distances let full fault tolerance run at $d = 9$–13 instead of 15 at $p = 10^{-4}$. Either this change or optimized lanes turns the only cell in which STAR led alone ($p = 10^{-4}$, $\epsilon = 0.01$, $Q = 10^5$) into a tie. No STAR-only window remains.

Raw RUS is a model option, not a conclusion. Within our instrument, averaging over RUS branches cancels the coherent term to first order. With success probability one half and conjugation on failure, the first-order coefficient of the mth retry's coherent term is $2^{-m-1} - \sum_{n>m} 2^{-n-1} = 0$, while dephasing keeps the weights 2^{-m}. Raw RUS then carries about half of noisy PCEC's dephasing; with the growth bound composed, the exposure falls from 0.685 to 0.452 at 4×4 and from 12.6 to 8.0

Change	Cells	Mechanism
FT delivery	8	Larger batches cut synthesized rotations; ports and factories cut the attempt time
Richardson	6	Base step count 8τ with a K_4/r^4 residual frees the Trotter allowance at $\epsilon = 0.001$
Lanes	4	Parallel analog injection, mainly for the hybrid
Distances	4	Full fault tolerance at $d < 15$ when $p = 10^{-4}$
Raw RUS	1	Lower STAR and hybrid exposure within the restricted instrument
All five	15	Mostly earlier separation of full fault tolerance

Table 2: Cells, out of 64, in which each change alters the encoded capacity leader relative to the reference menu.

at 16×16. Toshio *et al.* [4] average the same branches in their Appendix E, yet report a raw-RUS prefactor slightly larger than with PCEC and a logarithmic angle dependence. We have not reconciled the two. Raw RUS changes the leader in only one cell, so the conclusions about full fault tolerance do not depend on it. The STAR and hybrid gains in Fig. 1 do; for example, 16×16 STAR at $\tau = 1$ and $\epsilon = 0.01$ falls from $10^{10.1}$ to $10^{8.0}$ qubits with raw RUS alone.

The surface-code law is calibrated but not yet propagated. Both encoded arms use $p_L(d) = A(p/p_{\mathrm{th}})^{(d+1)/2}$ per patch per round with $A = 0.1$ and $p_{\mathrm{th}} = 0.01$, a suppression factor $\Lambda = 10$ at $p = 10^{-3}$. A circuit-level memory calibration under our noise model and 240 ns schedule, using Stim and PyMatching [5, 6] with 7.8×10^7 training and 1.8×10^7 holdout shots at $d = 3$–9, gives $\Lambda = 7.41$ (95% profile interval 7.20–7.62) and a summed X/Z amplitude of 0.077 at $p = 10^{-3}$. A fit restricted to $d \geq 5$ gives $\Lambda = 7.97$, and a single law is statistically rejected across $d = 3$–9, so the exponential is an approximate resource model. At $p = 10^{-4}$, $\Lambda = 89$ with a wide interval (64–146). The fitted law raises the required distance by 4–6 at $p = 10^{-3}$, or 1.3–1.5 times the patch area, and by at most 2 at $p = 10^{-4}$. Because STAR and full fault tolerance share the law, it creates no arm-specific advantage, although their different volumes and distance rounding can still shift relative costs. It does move both encoded arms relative to physical-qubit computation by roughly the area factor. The curves in this section use the reference law; surgery, correlated noise, and distances above 9 remain uncalibrated.

The fixed-time benchmark in Fig. 3 shows the same ordering with larger analog gains: at $p = 10^{-3}$, $\epsilon = 0.01$, and 10^8 qubits, STAR reaches $L = 16$ and the hybrid $L = 24$, compared with 12 and 16 under the reference menu. The conclusions remain conditional on the declared lane, port, and transfer charges, on the transfer of the finite-patch Trotter envelope beyond $\tau = 4$ and beyond 4×4, on the reference surface-code law discussed below, and on the assumed NISQ mitigation overhead. For the time-growing workload, full fault tolerance leads alone in every cell with at least 10^7 qubits and in three of four cells at 10^6, where the fourth is a tie with the hybrid. The analog arms pay per radian and full fault tolerance per logarithm; in this workload the radians grow faster.

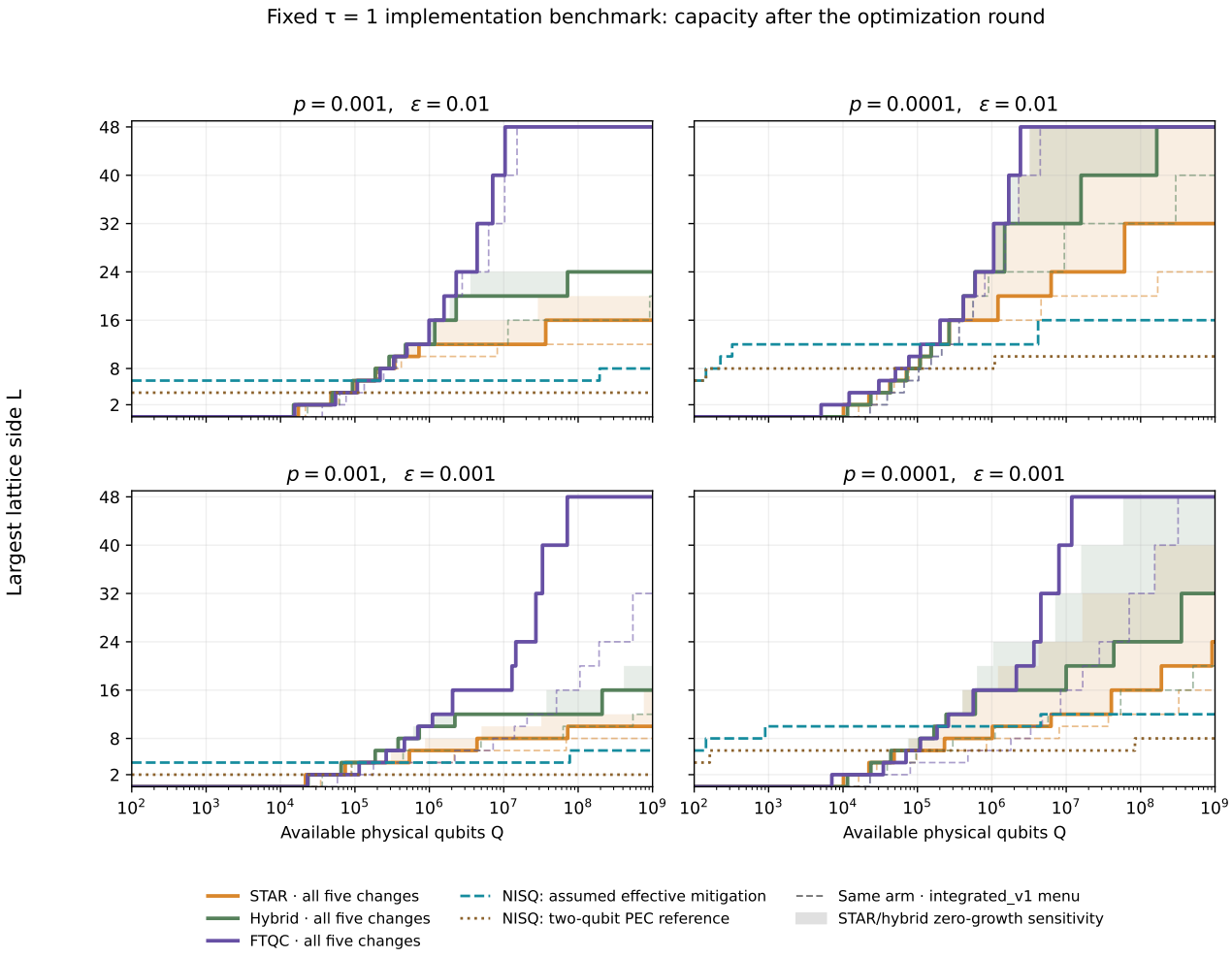


Figure 3: Fixed-time benchmark at $\tau = 1$, drawn as in Fig. 1. Beyond $L \approx 8$ these curves compare the cost of evolving full lattices, not the accuracy of a local answer.

2 Integrated comparison and accuracy requirements

Section 1 supersedes the capacity ordering of this section by optimizing five of its fixed choices; this section defines the reference menu that it reproduces. The central question is which hardware and algorithm assumptions determine the least expensive way to obtain a useful observable. We lead with absolute error $\epsilon = 0.01$ and keep $\epsilon = 0.001$ equally visible. The current comparison in Figure 4 combines the existing component evidence with the common resource accounting. It supersedes the older scalar-coefficient capacity figure as our discussion result. Earlier calculations remain below as controlled comparisons and reproducible historical references.

At $p = 10^{-3}$ and $Q = 10^6$, the central scenario reaches lattice sides 10, 12, and 12 for STAR, hybrid, and FT at $\epsilon = 0.01$, respectively. At $\epsilon = 0.001$, the sides are 4, 6, and 4. The scenario ranges overlap at these budgets. At $Q = 10^7$ and loose precision, FT reaches side 32 while STAR and hybrid reach side 12; FT also leads across the declared scenario menu there. These are conditional finite-lattice comparisons, with the same accuracy-transfer and delivery limitations as the figure. Conversely, at $p = 10^{-4}$, $Q = 10^5$, and $\epsilon = 0.01$, STAR reaches side 8 while hybrid and FT reach side 6 throughout this scenario menu.

Injection error and acceptance are coupled. Replacing the old coefficient $\alpha = 1.5$ with a larger constant would omit the dependence of preparation cost on angle and injection distance. Instead, we evaluate the existing restricted instrument at every distinct workload angle, including noisy probabilistic coherent error cancellation (PCEC) and repeat-until-success (RUS) corrections. For transverse channel multiplier c_j, the dephasing inverse contributes

$$\chi = -2 \sum_j \log(\operatorname{Re} c_j), \qquad b_{\mathrm{coh}} \le \frac{1}{2} \left[\prod_j \left(1 + \frac{|\operatorname{Im} c_j|}{\operatorname{Re} c_j} \right) - 1 \right]. \tag{3}$$

Rotation multiplicities enter the sums and products. The coherent term and finite RUS tail consume explicit error allowance, in addition to the existing reserve for unmodeled effects. This is a generator-fault submodel with ideal syndrome projection; detector rejection is still a declared model.

We consider injection seeds 5, 9, and 25, with measured growth anchors at distances 13 and 25 where available. The central scenario inserts each seed's simultaneous X/Z campaign upper limit as an independent dephasing component during growth. Those experiments include memory and readout and start with an ideally encoded state; their composition with arbitrary-angle injection is an assumption. Seed 25 avoids growth and pays its own acceptance. The zero-growth sensitivity is available for seed 9, whose zero observed failures constrain an upper limit rather than establish zero error. We make no inference about the untested distance-7 seed.

When the data distance exceeds the growth anchor, we stretch the reference service time by their distance ratio and charge a further protected handoff. The extra time and logical volume are explicit, but the corresponding channel has not been measured. The distance-13 receiver is included to avoid imposing the distance-25 area at low physical error. Analog parallelism retains the four-lane aggregate assumption. Cultivation quality and source choices are shared by FT and hybrid, including the slower-source scenario.

Synthesis is selected before comparing architectures. The FT menu includes individual synthesis and Hamming-weight batches 4, 8, and 16, with workspace and measurement-assisted uncomputation charged. Compact and fast blocks each have one T-consumption port. The checked arithmetic savings enter the comparison, while routing remains an aggregate assumption. The measured slowdown of one explicit onsite transport schedule is not treated as a full-program correction. Thus changes to the preferred batch or architecture remain possible when complete schedules are available.

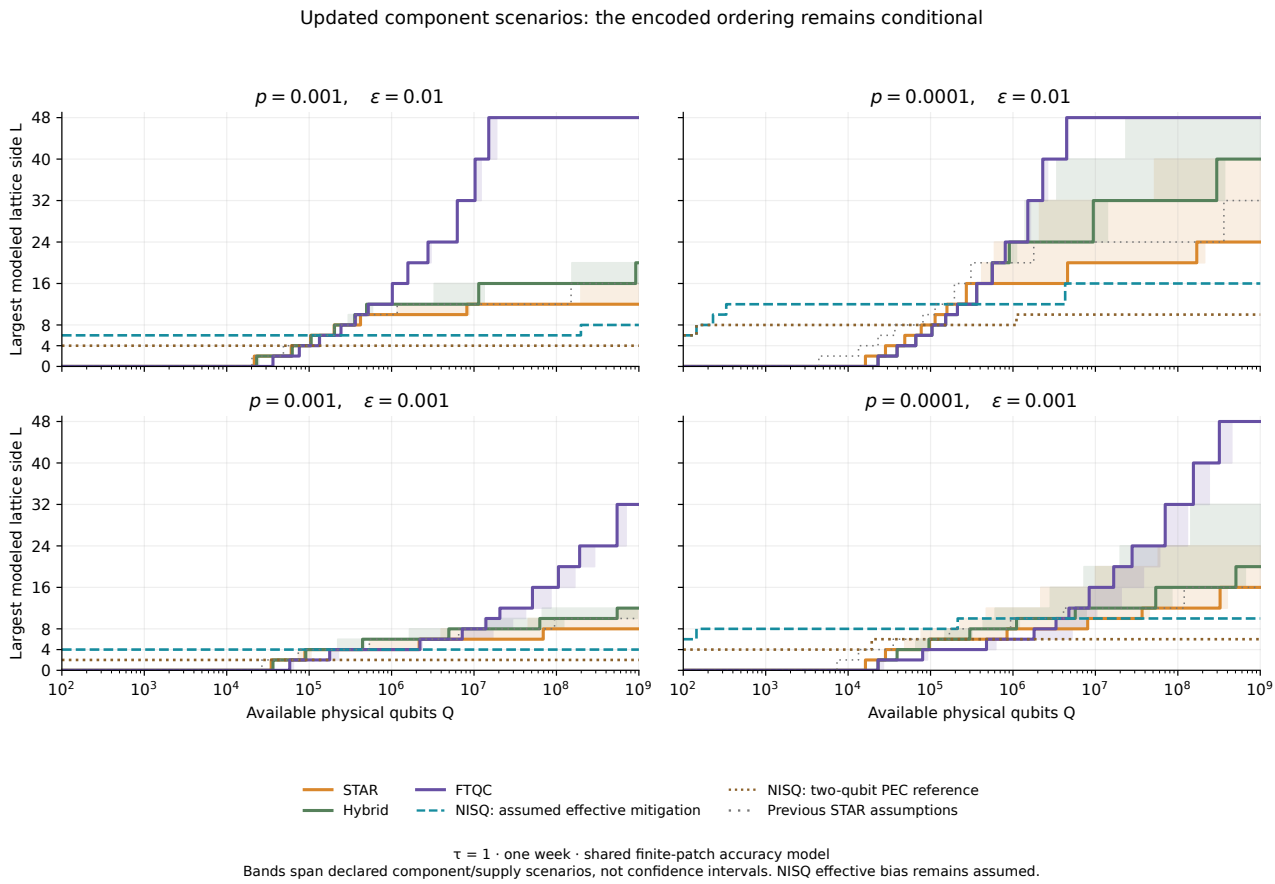


Figure 4: Current component-informed capacity at $\tau = 1$ and a one-week cutoff. STAR and hybrid use angle-dependent generator-fault channels with noisy PCEC, preparation acceptance, RUS retries, and explicit growth scenarios. FT selects individual or Hamming-weight synthesis and compact or fast one-port blocks. Solid encoded curves use growth-campaign upper limits as an independent component in the analog instrument. Bands include a zero-growth sensitivity for seed 9 and a slower rejection/supply scenario; they are not confidence intervals. The previous STAR line is shown for comparison. The NISQ effective line still assumes that its residual-error reserve is achievable, while the two-qubit PEC line describes the specified inverse-channel reference. Larger-lattice Trotter transfer and aggregate delivery remain assumptions; $L = 48$ is the sampled ceiling.

NISQ accuracy is a requirement to solve for. For a fixed physical estimator, let F be its variance per attempted circuit relative to $v_* = D_*(1 - D_*) = 0.09$, and let b be its remaining systematic offset. Rejection is included in F. At footprint a, attempt duration t_a, and deadline T, the evaluation budget after training is

$$M = \max\left\{0, \left\lfloor \frac{Q}{a} \right\rfloor \left\lfloor \frac{T}{t_a} \right\rfloor - M_{\text{train}}\right\}. \tag{4}$$

The Gaussian planning condition and its two equivalent requirements are

$$e_T + b + z\sqrt{Fv_*/M} \le \epsilon, \tag{5}$$

$$F_{\max}(b, r) = \frac{M(\epsilon - e_T - b)^2}{z^2 v_*}, \qquad b_{\max}(F, r) = \epsilon - e_T - z\sqrt{Fv_*/M}, \tag{6}$$

where $z = 1.96$ and $e_T = C/r^2$. An exhausted allowance is infeasible. Figure 5 optimizes over a declared integer-step menu; the estimator must achieve its F and b at the selected r. We do not assume that its overhead is independent of refinement.

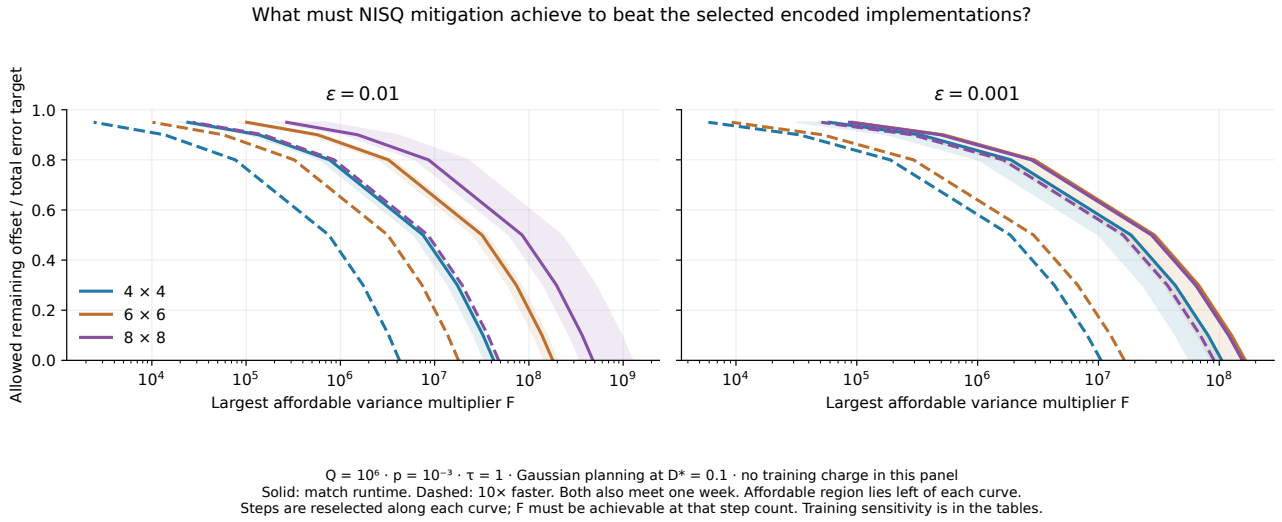


Figure 5: Requirements for physical-qubit computation at $Q = 10^6$, $p = 10^{-3}$, and $\tau = 1$. Each curve gives the largest affordable variance multiplier as a function of the allowed systematic offset. Solid curves match the fastest fitting member of the selected encoded implementations; dashed curves require a tenfold speedup. Both also meet the week cutoff. Bands span the encoded scenarios, with steps reselected along each curve. The panel assumes zero training charge; saved tables also charge six million training attempts. The affordable region lies to the left of a curve. These are Gaussian planning requirements on an unknown estimator, not predictions of an experimental pipeline's accuracy.

For a requested speedup s, we set $T = \min(T_{\text{week}}, T_{\text{encoded}}/s)$, with $s = 1$ or 10. The encoded comparator is the fastest fitting member of the three selected implementations within that scenario. Those implementations were optimized for one-week area, not reoptimized over all allocations for runtime at this particular Q. If no encoded copy fits, only the one-week physical requirement is reported. Training uses zero or six million additional equal-duration attempts as explicit sensitivities. A real pipeline must also account for uncertainty in its trained correction and any shared time-series information.

For example, at 4×4, $Q = 10^6$, and $p = 10^{-3}$, allowing a remaining offset $b = 0.001$ at $\epsilon = 0.01$ gives $F_{\max} = 3.3 \times 10^7$ at 13 steps, with a 67-second encoded comparator. At $\epsilon = 0.001$, allowing $b = 10^{-4}$ gives $F_{\max} = 8.1 \times 10^7$ at 40 steps, with a 14-hour comparator. These central-scenario

examples charge no training. The larger affordable multiplier in the strict case follows from the longer encoded runtime; it does not make physical accuracy easier. An actual estimator must attain both the stated offset and variance at the corresponding circuit depth.

The scientific decision is whether plausible accuracy and sampling costs fall on one side of these boundaries. A detailed mitigation-transfer calculation is most useful where resolving that question could change the ranking. We apply the same distinction between measured components and assumed composition to all three architectures. A capacity leader is called robust only when its worst capacity across this scenario menu exceeds the best capacity of both other encoded arms. Overlaps remain unresolved, and this criterion establishes no unconditional NISQ size boundary.

3 Checks against published operating points

Published demonstrations and resource estimates test different parts of the comparison. We first retain their tasks, estimators and hardware assumptions. This distinguishes an accounting discrepancy from a change in the scientific question. In particular, an energy-estimation resource table and a measured time series do not share the local-observable error contract of Section 2.

3.1 The TopQAD STAR and surface-code ledger

Table IV of Chung et al. [7] estimates Hubbard ground-state energies to error $0.005L^2$. STAR uses quantum complex exponential least squares (QCELS), while FT uses phase estimation with plaquette Trotterization and Hamming-weight phasing. Suitable initial ground-state overlap is assumed. We transcribe all 16 STAR and 32 FT entries. The published distances, retry clock counts and FT T counts are inputs to this check.

The STAR footprint and the deepest-circuit rotation count are

$$Q_{\mathrm{STAR}} = (4L^2+1)2d_f^2, \qquad R_{\max} = 2[8L(L-1)+L^2]r_{\max}. \tag{7}$$

For $n = 1,\dots,5$ and $j = 1,\dots,J$, the QCELS time fraction is $f_{nj} = (n/5)2^{j-J}$ and we use $r_{nj} = \lceil r_{\max} f_{nj} \rceil$. With 100 samples for each of the real and imaginary quadratures, the PEC-free evolution-time ledger is

$$T_0 = 200(d_i + d_f + 6)t_{\mathrm{cyc}} \sum_{j=1}^{J}\sum_{n=1}^{5}\big[(C_{\mathrm{step}}+18)r_{nj}+16\big]. \tag{8}$$

Here C_{step} is the supplied retry-schedule clock count and the additional terms are the controlled-evolution overhead from Appendix D. The 350 ns syndrome round includes six 25 ns gate layers plus measurement and reset. The zero-time circuit is omitted from this evolution-time sum; a physical campaign also prepares and reads out its samples.

All 16 STAR areas reproduce the published two-significant-digit values. Fourteen of 16 PEC-free runtimes agree within 2.4%. For FT, consuming one T state per d_f syndrome rounds gives $T_{\mathrm{FT}} = N_T d_f t_{\mathrm{cyc}}$, which reproduces all 16 time-optimal entries within 2.8%. This checks consumption given the supplied T count; it does not reconstruct the private factory and layout optimizer.

The growth fit parameters and complete retry schedule have not been independently reproduced. We instead check their consistency across workloads. Using the paper's distance-9 small-angle coefficient β, the PEC exponent at each QCELS point is

$$\chi(f) = 4\beta\Theta_{\max} f + 8I_g R(f), \qquad \Theta_{\max} = \frac{T_{\max}}{2}\left[4L(L-1)+\frac{UL^2}{4}\right], \tag{9}$$

where $T_{\max} = 0.06/(2\epsilon/3)$ and I_g is the growth contribution per state. The factor eight combines the PEC factor four with two growth contributions per successful rotation. Fitting I_g only at $U = 4, L = 4$ gives 3.17×10^{-6} for target hardware. Carrying it unchanged to the other seven target-hardware entries reproduces their total runtimes within 10%. This is a calibrated transfer check, not a forward gadget prediction.

Two desired-hardware entries remain flagged. At $U = 8, L = 8$, the listed 750 clocks predict 470 ms rather than the listed 297 ms. At $U = 8, L = 6$, the reported total-to-PEC-free ratio is below the angle-only estimate under these conventions, so no nonnegative I_g reproduces it. Agreement in total runtime can conceal compensating differences in its components.

On the fixed target-hardware 4×4 workload, changing only the clock from 350 to 240 ns multiplies runtime by 0.686; doubling β multiplies it by 1.13. Substituting our distance-9-to-13 growth upper

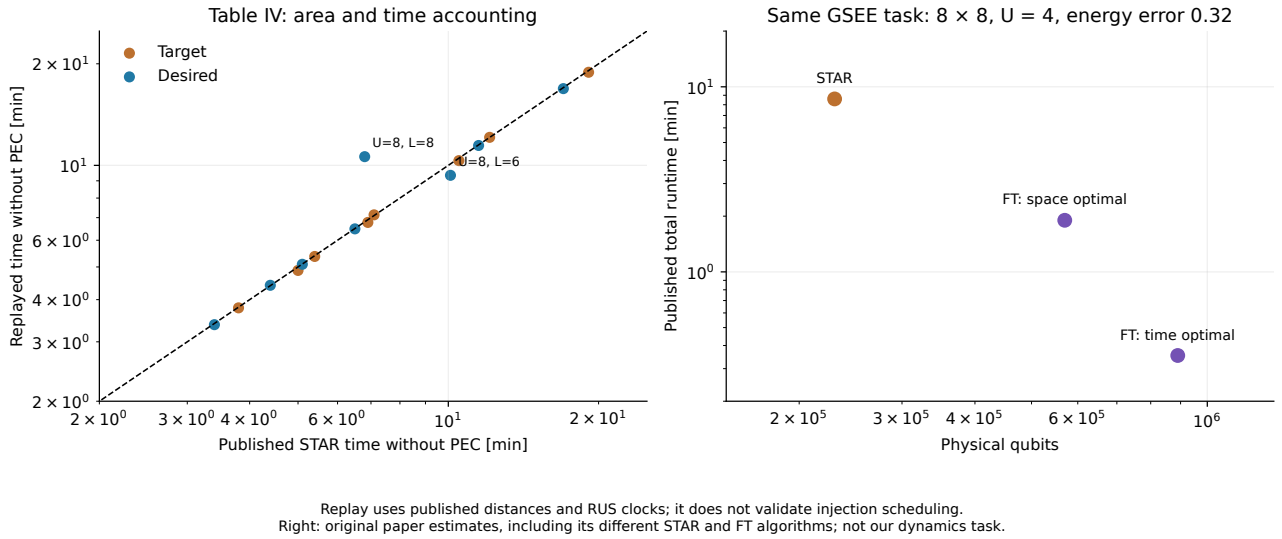


Figure 6: Published resource accounting on the original energy-estimation task. Left: STAR campaign times without PEC, reconstructed from the QCELS grid and supplied gadget clocks. Two desired-hardware entries remain discrepant. Right: the published 8×8, $U = 4$ target-hardware estimates show the area–time tradeoff: STAR uses 230,000 qubits for 8.6 minutes; FT uses 570,000 for 1.9 minutes or 890,000 for 21.2 seconds. These use the paper's different STAR and FT algorithms and are not predictions for our dynamics observable.

bound instead multiplies it by 5.81. The last substitution changes the noise and protocol interpretation and is only a sensitivity. It identifies growth as the larger unresolved input at this point, rather than providing a calibrated translation between the two hardware models.

3.2 The Phasecraft/Quantinuum experiment

We reduce the public shot archive and supplied processed observables for the 56-qubit experiment of Alam et al. [8, 9]. Its 7×4 periodic lattice has π flux and a dimer/defect initial state, evolved for four Trotter steps. The archive contains 160 shots at each of 20 times for each of $U = 0$ and $U = 4$. Direct reconstruction from the occupation bits reproduces every saved raw mean doublon to floating-point precision. We retain all 28 local doublons and the spatial mean; no trained correction or Gaussian-process smoothing is refitted.

For a shot with occupations $n_{i\sigma}$, the spatial estimator is

$$\overline{D} = \frac{1}{28}\sum_i n_{i\uparrow}n_{i\downarrow}, \qquad \mathrm{Var}(\overline{D}) = \frac{1}{28^2}\sum_{i,k}\mathrm{Cov}(D_i, D_k). \tag{10}$$

At $U = 4$, the median raw variance per shot is 0.0035, compared with 0.149 for the mean individual-site Bernoulli variance and our planning reference $v_* = 0.09$. Spatial averaging and correlations thus change the shot budget substantially. Training and smoothing additionally share information across circuits and times. The shot arrays do not identify pseudo-twirling groups, so our pooled variance is a diagnostic; the figure retains the authors' reported uncertainties, including their treatment of twirling and processing.

For the spatial mean at $U = 0$, the RMS difference from the FLO reference drops from 0.0259 raw to 0.0115 with TFLO and 0.0064 after smoothing. At $U = 4$, the corresponding differences from the bond-2048 TDVP reference are 0.0584, 0.0378 and 0.0362. The latter differences combine experimental,

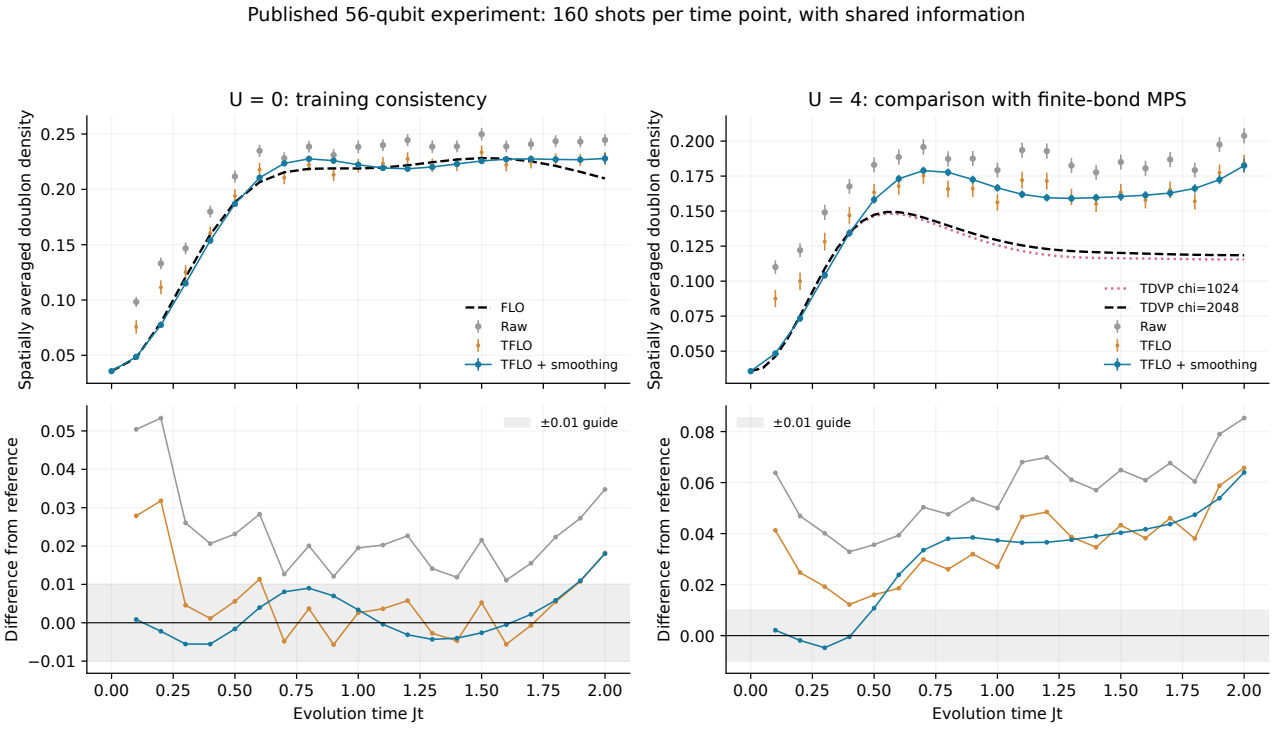


Figure 7: Published experimental estimates and classical references, reduced from the pinned data archive. All 20 experimental times are shown. The $U = 0$ comparison uses training information and is a consistency check. At $U = 4$, the TDVP curves are finite-bond MPS references, not exact truth; differences also contain Trotter error and cannot be identified with mitigation bias. The ± 0.01 band is a visual guide, not an accuracy certificate. Error bars are the published standard errors.

Trotter and reference errors. For the processed local doublons, the median reported standard error is 0.0146, compared with 0.0028 for the spatial mean. The useful low-shot experimental curves therefore do not imply the distinct 0.001 local-observable contract, while a gatewise PEC budget for that contract does not predict the experiment's shot count.

These checks establish consistency of published operating points with the accounting and identify the inputs still requiring prediction: observable accuracy for a chosen physical estimator, full noisy injection and growth, and complete magic-state delivery. They do not reduce the architecture scenario bands to statistical error bars. We next examine the superconducting data, whose geometry is closer to our baseline.

3.3 Willow: observed accuracy and the sampling design

The public Willow data [10, 11] permit a direct check across circuit depths and lattice sizes. Our declared selection contains all center-hole cases at 4×4 and 6×6, with zero or π flux, $U = 0, 4, 8$, and one to three Trotter steps. Each of the 36 cases has 24 nonzero evolution times and 200,000 attempted shots per time, organized as 20 randomized compilations of 10,000 shots. The reduction therefore covers 172.8 million attempted shots. These are reused public measurements, not new quantum runs.

The physical footprint is 34 or 74 qubits, including two ancillas, while 32 or 72 modes encode the fermions. Table I of the paper gives $G_2 = 400r + 20$ at 4×4 and $G_2 = 1440r + 52$ at 6×6. At three steps these are 1220 and 4372 two-qubit gates. The circuit scale is consistent with our native-gate proxy. Its estimator, however, uses selection, training, symmetry and shared time-series information. A single time series costs 4.8 million attempted shots, with an additional $U = 0$ training series whose cost can be shared by interacting targets.

For $U = 4$, zero flux and three steps, processed local doublons differ from the bond-2048 reference by 0.0096 RMS at 4×4 and 0.0479 at 6×6. For the spatial means the differences are 0.0068 and 0.0451. The median reported standard errors of the constituent local Z estimates are 0.0156 and 0.0810, respectively. These observed operating points help explain a useful physical-qubit calculation with tens of qubits, while leaving the stricter accuracy requirement in our architecture comparison open. The interacting exact-reference groups are empty in this archive, despite the paper showing exact 4×4 comparisons. We therefore retain the approximate-reference qualification even at 4×4.

Two archive conventions affect the comparison. First, values labeled `raw` already include number selection and diagonal symmetry averaging. The randomized-compilation means must receive equal weight after rejection; pooling all accepted shots gives different weights. We reproduce these raw means in all 36 workloads, with number-deviation thresholds of zero at 4×4, two at zero-flux 6×6, and one at π-flux 6×6. Second, the $U = 0$ `exact` values change with step count and describe the free circuit, not continuous dynamics. We compute a separate continuous FLO reference by orbital evolution, checked against the existing many-body sector solver. The supplied initial-state JSON is inconsistent with the center-hole labels; we instead identify and check the product input from its zero-time reference occupations. These distinctions are recorded in the data audit.

The processed archive supplies marginal Z and ZZ errors, without their cross-covariances. For a derived observable $O = \sum_i c_i O_i$, we retain the covariance-independent upper bound

$$\mathrm{SE}(O) \leq \sum_i |c_i| \, \mathrm{SE}(O_i). \tag{11}$$

We do not infer independent sites or treat this statistical bound as a bound on total error. The figures show the derived central values rather than presenting that loose bound as a measured confidence interval.

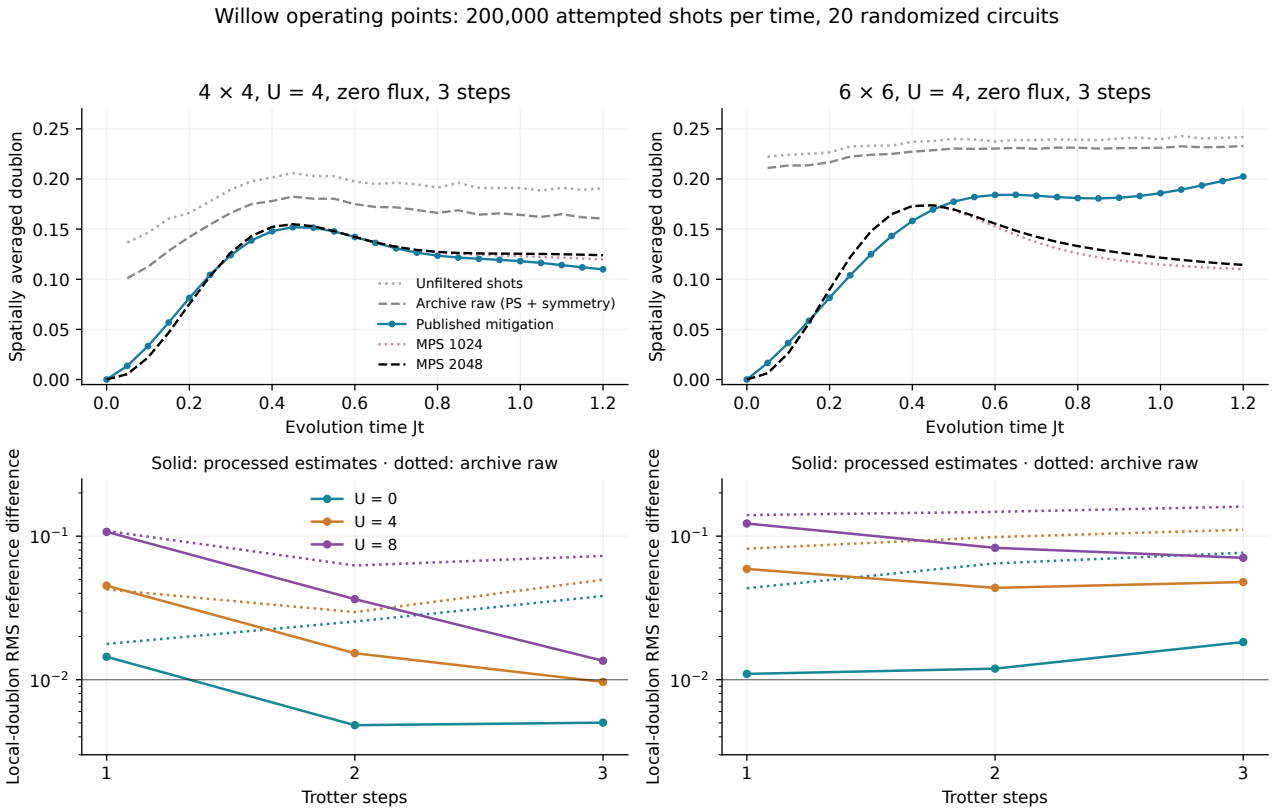


Figure 8: Willow experimental operating points. Upper panels show the spatial mean doublon at $U = 4$, zero flux and three steps, with finite-bond MPS references. Lower panels compare all local doublons at all 24 measured times as the step count changes. The $U = 0$ reference is independent continuous free-fermion evolution, but the experimental estimator uses $U = 0$ training. The interacting references are approximate MPS calculations. Differences are not certified experimental errors, and the 0.01 line is a guide.

Additional sampling must specify what is repeated. Shots of one randomized circuit share its noise realization. Let K be the number of independent randomized circuits, a the variance of their conditional expectations, and v/n the typical variance of each conditional sample mean. A useful planning decomposition is

$$\mathrm{Var}(\widehat{O}) \simeq \frac{a + v/n}{K}. \tag{12}$$

Increasing n at fixed K leaves the a/K contribution. Increasing the number of independent circuit realizations reduces both contributions. We estimate them from the 20 recorded groups, retaining acceptance, and separate that planning extrapolation from the directly measured variance of their mean. The inferred a is uncertain and is not an irreducible hardware error floor.

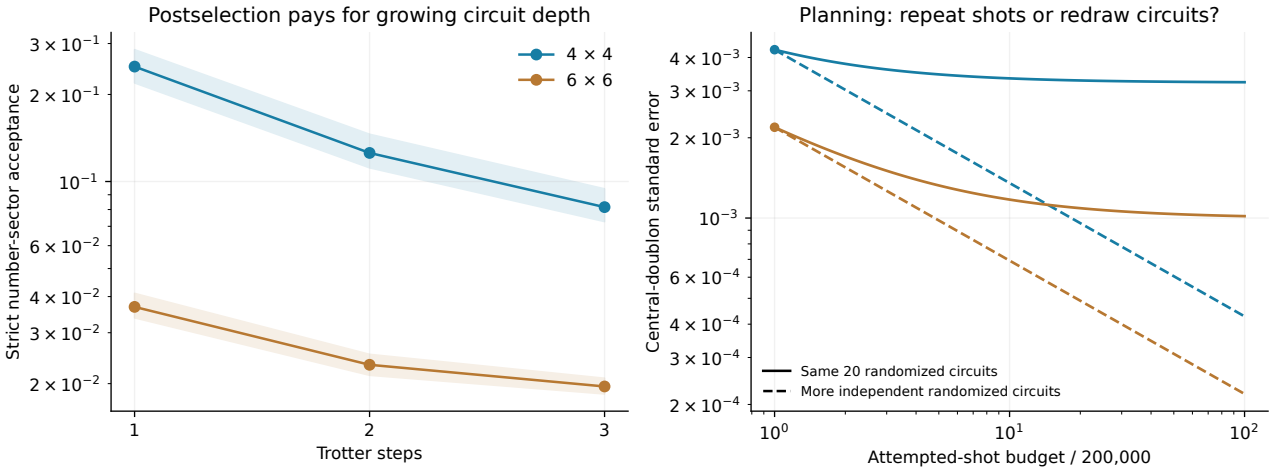


Figure 9: Measured acceptance and a sampling-design sensitivity. Left: strict number-sector acceptance, with median and range across the declared times, fluxes and interactions. Right: the central doublon's sampling uncertainty at $U = 4$, zero flux, three steps and $Jt = 1$, before TFLO and smoothing. The right panel retains the corresponding published number filter: exact sector at 4×4 and up to two total number deviations at 6×6. Dots show standard errors across the observed circuit groups. Curves extrapolate Equation (12), assuming independent new groups with the same distribution. They do not predict mitigation accuracy.

The experimental comparison therefore constrains the physical arm at two levels: a directly measured circuit and sampling cost, and an observable-dependent comparison with available references. It supports useful small-footprint dynamics without certifying every local observable at 0.01 or 0.001. We now calculate a matched interacting reference for the main 4×4 example.

3.4 Separating circuit error from experimental discrepancy

For the 4×4, $U = 4$, zero-flux center-hole input, we compute both continuous Hamiltonian dynamics and the ideal experimental circuits at all 24 saved times and all three step counts. This input has seven up spins, eight down spins and a hole at $(1, 1)$; it differs from the half-filled Néel input in our resource baseline. Its fixed-particle sector contains $\binom{16}{7}\binom{16}{8} = 147232800$ amplitudes.

The circuit uses the hopping order specified by the swap network in Appendix D2 of Ref. [10]. In a fixed fermionic frame, each step is an ordered half sweep of hopping terms, all onsite rotations, and the reverse hopping half sweep. Independent orbital evolution reproduces the archived $U = 0$ circuit

references at all 288 saved combinations of size, flux, step count and time. This check requires the archive's diagonal symmetry averaging: the maximum local-Z discrepancy falls from 0.345 without averaging to below 6×10^{-15} with it. We apply the same averaging to the interacting circuit observables. For this input and zero-flux Hamiltonian, diagonal reflection leaves the continuous target unchanged. The ordered finite-step circuit can break that symmetry, so applying the averaging to its reference matters.

Continuous evolution uses matrix-free Chebyshev propagation in increments of 0.05, with a conservative spectral bound of 64. We retain the analytical state-truncation bound and check floating-point behavior through full-state comparisons with independent small-system Fock-space calculations and a separate direct propagation to $Jt = 1$. Ideal circuits use exact two-mode hopping and onsite phase updates. These are numerical references for the specified finite-system unitary, not noisy hardware simulations.

Let D_H denote continuous dynamics, D_C the ideal circuit result, and $\widehat{D}$ the published experimental estimate. Then

$$\widehat{D} - D_H = (\widehat{D} - D_C) + (D_C - D_H). \tag{13}$$

The first contribution contains finite sampling and experimental processing effects. The second is deterministic circuit error. Squaring and averaging therefore retains a cross term; the two RMS differences cannot be added or interpreted as independent error bars. We report observed discrepancies without identifying them with the ensemble bias of an estimator.

The two contributions respond oppositely to refinement:

Steps	Experiment–circuit RMS	Circuit–continuous RMS	Total RMS
1	0.00469	0.04481	0.04429
2	0.00764	0.01973	0.01581
3	0.00946	0.01315	0.01030

More steps improve the ideal dynamics while the observed circuit discrepancy grows. At three steps the total RMS is smaller than the circuit error alone because the signed contributions partially cancel. This is a measured depth–accuracy tradeoff in the tested experimental pipeline, not a projection of its performance beyond three steps.

At three steps, 71.4% of the 384 local time-site points lie within 0.01, while the largest difference is 0.0400. The bond-2048 MPS reference differs from continuous dynamics by 0.00214 RMS and at most 0.00909. Thus its previous use gave a reasonable overall accuracy scale, but could not resolve a 0.001 comparison. These statements concern the saved data set; they are not statistical coverage guarantees.

This comparison tests useful dynamics on the experiment's actual small device. It neither substitutes a fitted mitigation exponent into the capacity model nor transfers this input's Trotter coefficient to the Néel baseline. The distinction between a roughly 0.01 RMS experimental result and a specified local-observable error budget remains central to comparing the physical and encoded architectures.

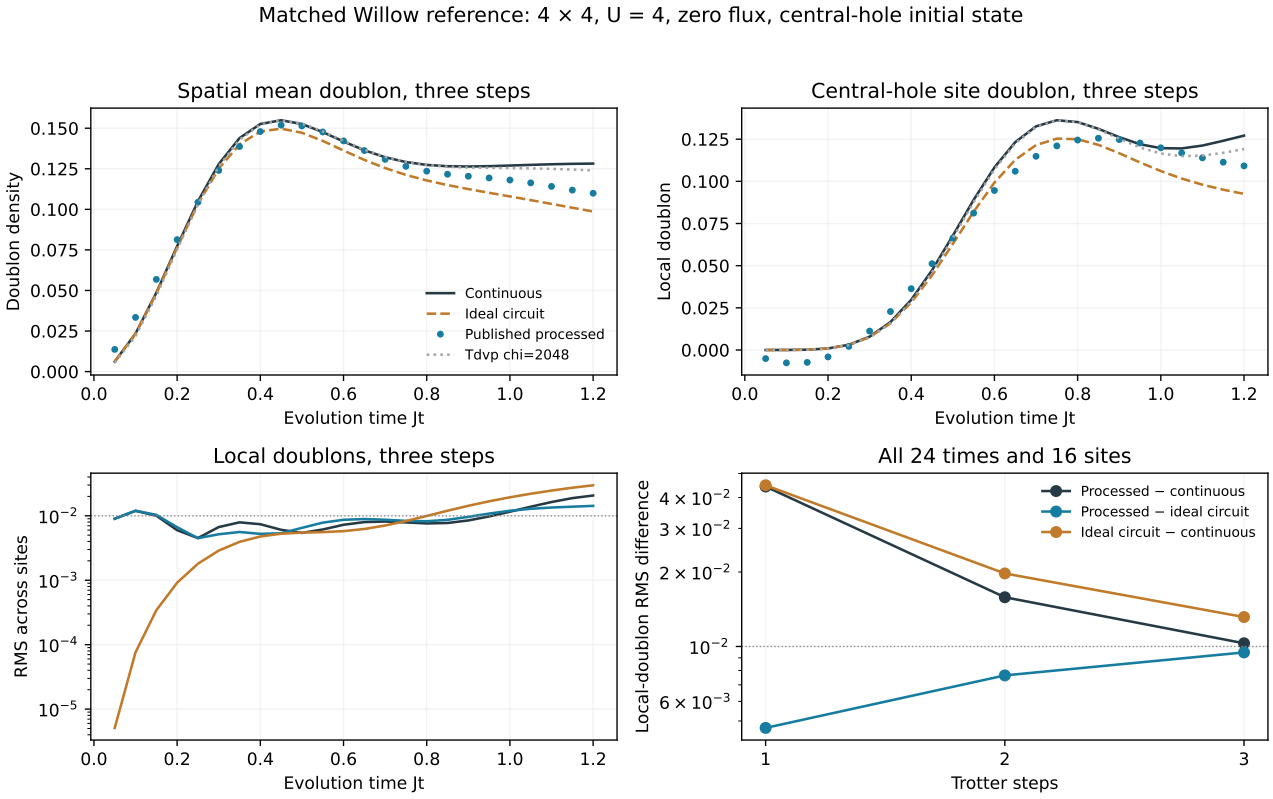


Figure 10: Matched interacting reference for the 4×4, $U = 4$, zero-flux center-hole experiment. Upper panels: spatial mean and central-site doublons at three steps. Lower left: local-doublon RMS differences across sites at each time. Lower right: RMS differences over all 24 times and 16 sites versus step count. The ideal circuit includes the same diagonal averaging as the processed estimates. Signed discrepancies can reinforce or cancel, so the three RMS curves do not add. The experimental curves retain finite sampling and processing error; no larger-lattice transfer is assumed.

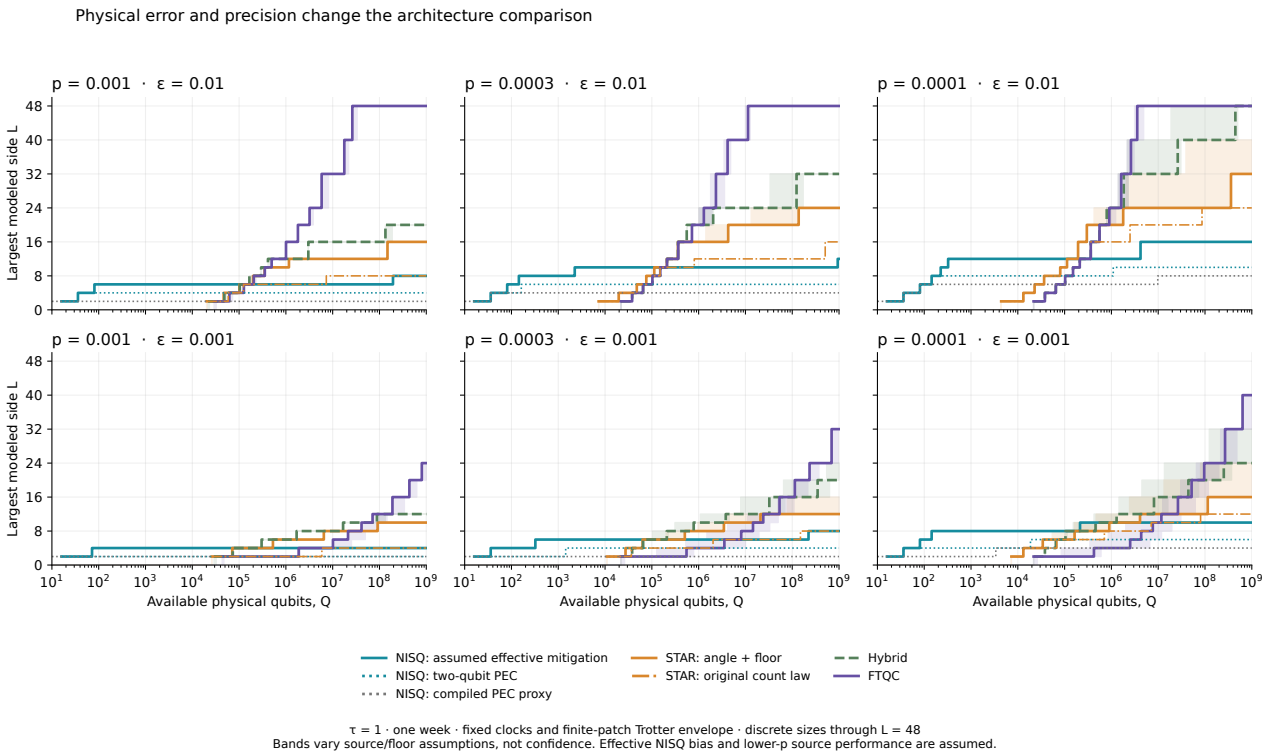


Figure 11: Achievable lattice side versus available physical qubits at $\tau = 1$ and a one-week cutoff. Columns vary physical error; rows show 0.01 and 0.001 absolute error. Solid encoded curves retain the source and injection floor at their $p = 10^{-3}$ reference values. Bands span the explicit source/floor scenarios described in the text, not statistical confidence intervals. The count-law STAR curve changes the injection exposure while retaining the same delivery model. NISQ retains separate assumed-effective, two-qubit PEC, and compiled-PEC scenarios. Only the sampled sizes are shown; $L = 48$ is the sweep ceiling. Larger-lattice accuracy transfer and effective mitigation bias remain assumptions.

4 Earlier physical-error and algorithm sensitivities

This section retains the previous scalar-coefficient comparison. Its numbers are historical controlled sensitivities; Section 2 is the current integrated result. We lead the workshop example with 0.01 absolute error on the local doublon observable and retain 0.001 as a second principal target. Figure 11 varies p from 10^{-3} to 10^{-4} while keeping the clocks, Hamiltonian, initial state, observable, and finite-patch Trotter model fixed. The $p = 10^{-3}$ central rows reproduce all 144 corresponding entries of the previous added-patch comparison. The new sweep contains 864 scenario points, with per-arm error allocation and source counts reoptimized. The model remains a finite-lattice workload comparison: at fixed time, an independently justified local-patch approximation could change what increasing L means.

Source assumptions remain visible. We do not infer a new cultivation protocol from a change in p. The held scenario keeps $q_T = 2 \times 10^{-9}$, a 1024-qubit source, 100 preparation cycles, and the distance-15 handoff constraint. A second scenario assumes $q_T(p) = 2 \times 10^{-9}(p/10^{-3})^3$ and an injection floor $f(p) = 10^{-5}(p/10^{-3})$, with source area and cycles unchanged. A third keeps both error floors fixed and multiplies source preparation cycles by four. These choices bracket specified sensitivities; they are not calibrated lower-p implementations or guaranteed bounds on an optimal source. In

particular, the inherited handoff convention imposes $d \geq 15$ on the FT/hybrid data block as well, so its low-p area floor is a model choice. The ideal cleanup term $35q_T^3$ accounts for input-state errors. Faults in the cleanup block and waiting storage are charged separately through the protected tile-time volume and logical-error budget.

At $Q = 10^6$ and $\epsilon = 0.01$, lowering p from 10^{-3} to 10^{-4} raises the held-scenario STAR capacity from 10×10 to 20×20, and FT/hybrid capacity from 12×12 to 24×24. The NISQ effective and two-qubit PEC capacities increase from sides 6 and 4 to 12 and 8, respectively. At $\epsilon = 0.001$, the corresponding STAR and FT sides are 6 and 2 at $p = 10^{-3}$, and 10 and 4 at $p = 10^{-4}$. The latter comparison is sensitive to the magic-state assumption: for the strict 4×4 FT task at $p = 10^{-4}$, holding q_T fixed costs 4.23×10^5 qubits, whereas the assumed cubic improvement gives 8.47×10^4. Lower physical error benefits every arm, but the relative window cannot be specified by p without the source assumptions.

Injection law as a controlled switch. For the original injection model [12], we use the exact dephasing-inverse exposure

$$\chi_{\text{count}} = -2R \log(1 - 8p/15), \qquad \chi_{\text{angle}} = 4\alpha p\Theta + 4fR. \tag{14}$$

Both calculations use the same surface-code law and delivery timing, so this switch isolates the exposure law rather than reconstructing two complete physical protocols. At $p = 10^{-3}$, the optimized strict 4×4 STAR estimate changes from 7.14×10^4 qubits under the angle law to 5.87×10^6 under the count law. At $p = 10^{-4}$, both estimates fit in one 13,122-qubit copy. Thus a change in the injection law can dominate the strict comparison at the original noise point without producing a fixed architecture advantage at all p.

Repeated-angle synthesis changes the FT comparison. We implement Campbell's Hamming-weight phasing construction for equal-angle rotations within each plaquette stage [13]. A batch of b rotations uses

$$a_b = b - \text{popcount}(b), \qquad k_b = \lfloor \log_2 b \rfloor + 1 \tag{15}$$

clean ancillas/measurement-assisted Toffolis and synthesized rotations, respectively. We charge four T states per Toffoli, synthesize all k_b weight-register rotations within the common error budget, and reuse the a_b ancillas between batches. The peak workspace is charged for the full shot; arithmetic adds two logical Clifford clocks per Toffoli to the reference timing model. We retain all fixed plaquette T gates and do not pool equal angles from different times. Batch size one exactly reproduces individual synthesis. The nearest-neighbor surgery layout and effective delivery ports remain uncompiled reference assumptions.

At $p = 10^{-3}$ and $\epsilon = 0.001$, batch size 16 reduces the 4×4 estimate from 1.88×10^6 to 2.78×10^5 qubits. The optimized circuits consume 192,332 and 80,123 T states per shot, with 33 and 36 steps, respectively; the latter uses 15 extra logical ancillas. The area improvements at $L = 8$ and 16 are factors 5.78 and 6.04, using batches of 64 and 63 live ancillas. At $\epsilon = 0.01$, individual synthesis remains cheaper at $L = 4$ and 8 because one copy already completes the campaign and the extra workspace raises its area. These results change the strict FT comparison substantially without changing the Hamiltonian simulation algorithm. The principal p sweep deliberately retains individual synthesis to preserve its frozen baseline.

At strict 4×4, neither selected circuit uses a cleanup stage. The lower T demand reduces the magic-state error allocation from 38.5% to 16.0% and increases the statistical allocation from 29.8% to 58.0%. The reduced shot count explains why the area gain exceeds the reduction in T count: repeated-angle synthesis changes the optimal estimator budget as well as the cost of a circuit.

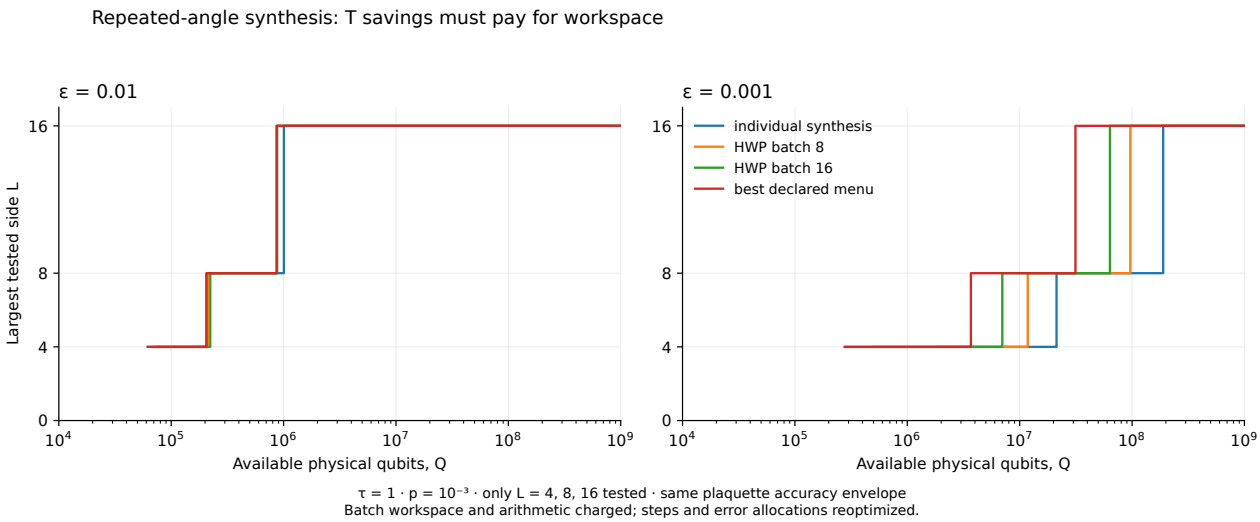


Figure 12: FT capacity with repeated-angle synthesis, for the three tested lattice sides 4, 8, and 16. The best-menu curve chooses individual synthesis or an allowed Hamming-weight batch, with steps and error budgets reoptimized. The circuit, plaquette accuracy envelope, source menu, and delivery convention are common. The staircase ends at the largest tested size; it is not an FT capacity plateau.

Local encoding and local ordering are different changes. The compact encoding maps hopping to Pauli operators of weight at most three and uses fewer than 1.5 qubits per fermionic mode on the square lattice [14]. We evaluate a $3N$-qubit register scenario, with four CNOTs per hopping Pauli and two per onsite Pauli, plus $8N$ assumed preparation gates. A two-CNOT hopping variant assumes fusion; a routing variant doubles movement and placement costs. Preparation, embedding of both spin sectors, and gate utilization are explicitly uncompiled assumptions. The generic parity gadgets use 6,754 two-qubit gates at $L = 4, r = 8$, compared with approximately 3,455 in the native JW proxy. The fused hypothesis uses 3,682. Removing asymptotic routing growth therefore does not guarantee savings at the small lattices accessible under the present noise assumptions.

For STAR, the compact scenario enlarges the data register and separately tests four and sixteen assumed rotation lanes. Four lanes cost more area than the JW reference at $L = 4$, and sixteen lanes cost still more because the selected calculation already fits in one copy. Locality does not automatically pay for parallel delivery. These resource scenarios retain the existing termwise error envelope; transferring it to larger lattices remains an assumption.

A separate exact-sector diagnostic isolates ordering. Keeping the Hamiltonian, initial state, JW representation, and original onsite split fixed, a four-color bond permutation changes the $\tau = 1$ coefficient $|D_r - D|r^2$ from approximately 0.0344 to 0.0463 on 2×2 and from 0.0766 to 0.0913 on 3×3. Grouping all onsite terms instead gives 0.2231 and 0.2178, matching the plaquette results on these patches. These measurements show that a more parallel ordering need not reduce the local-observable error. They neither establish bulk convergence nor replace the frozen accuracy envelope.

4.1 Qubitized dynamics for the same Hamiltonian

We also count a qubitized implementation of the open-boundary Hamiltonian, using pyLIQTR and Qualtran components with quantum singular value transformation (QSVT) [15, 16]. The specialized Hubbard encoding in the installed library has periodic boundaries; using it directly would change the

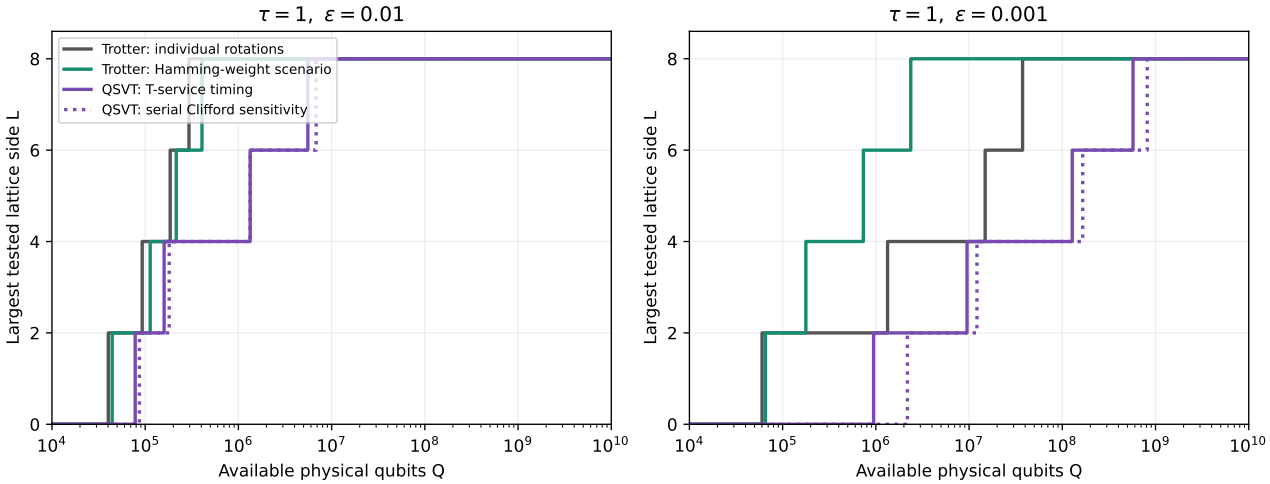


Figure 13: Same-task FT algorithm comparison at $p = 10^{-3}$ and one week. The vertical coordinate is the largest tested side among $L = 2, 4, 6, 8$; the plateau at eight is the test ceiling. Trotter uses the finite-patch accuracy envelope and optimized budgets, with individual synthesis or the aggregate Hamming-weight scenario. QSVT uses the cheaper of the repaired generic and equal-weight encodings, with state selection and amplification charged. The dotted curve adds Clifford serialization. These conditional costs do not include a fully routed SELECT or full Hamming-weight layout.

task. The centered open-boundary Pauli expansion instead has

$$\alpha = \sum_{\ell} |h_\ell| = 4L(L-1) + \frac{U}{4}L^2, \qquad \alpha\big|_{L=4,U=4} = 64. \tag{16}$$

At $U = 4$, hopping coefficients have magnitude $1/2$ and onsite coefficients have magnitude one. Repeating each onsite Pauli twice makes all LCU weights equal. There are $8L(L-1)+2L^2$ entries, exactly 128 at $L = 4$, so PREPARE requires only seven Hadamards there. The controlled block encoding costs 508 T gates and no arbitrary rotations. Generic amplitude preparation at the same point costs 508 T gates plus 256 arbitrary rotations. A local adapter places coefficient signs in SELECT and magnitudes in PREPARE; zero-weight padding selects the identity. The Hamiltonian is independently checked against full fermionic creation and annihilation operators on 2×2.

We use half-amplitude cosine and sine polynomials and combine their controlled branches, giving accepted amplitude $1/4$ before amplification. The accounting includes rejected attempts, zero through three oblivious amplification steps, polynomial and phase-realization error, rotation synthesis, conditional logical-fault bias, and the shared magic-state source menu. Appendix C gives the approximation budget. These are constructive polynomial and compiled-component counts; large-degree phase lists and routed SELECT circuits have not been validated. Both algorithms use the fast data block here. A QSVT sensitivity either omits extra Clifford serialization or charges all counted Cliffords plus two Clifford slots per T; neither choice is a physical nearest-neighbor SELECT schedule.

At $L = 4, \tau = 1, \epsilon = 0.001$, the selected QSVT circuit uses 551,403 T states per attempt and costs 9.51×10^6 physical qubits under the T-service timing assumption. Individual Trotter uses 192,332 T states and 1.34×10^6 qubits; the Hamming-weight scenario uses 82,336 T states and 1.77×10^5 qubits. At $\epsilon = 0.01$, the respective QSVT and individual-Trotter costs are 1.60×10^5 and 9.29×10^4 qubits. QSVT does not beat the better Trotter scenario at any of the 32 tested size, time, and precision points. This comparison does not exclude a more structured open-boundary encoding or a coherent observable estimator. The $\tau = 8$ Trotter comparison still uses only the 3×3 time envelope, so its transfer to larger lattices remains particularly uncertain.

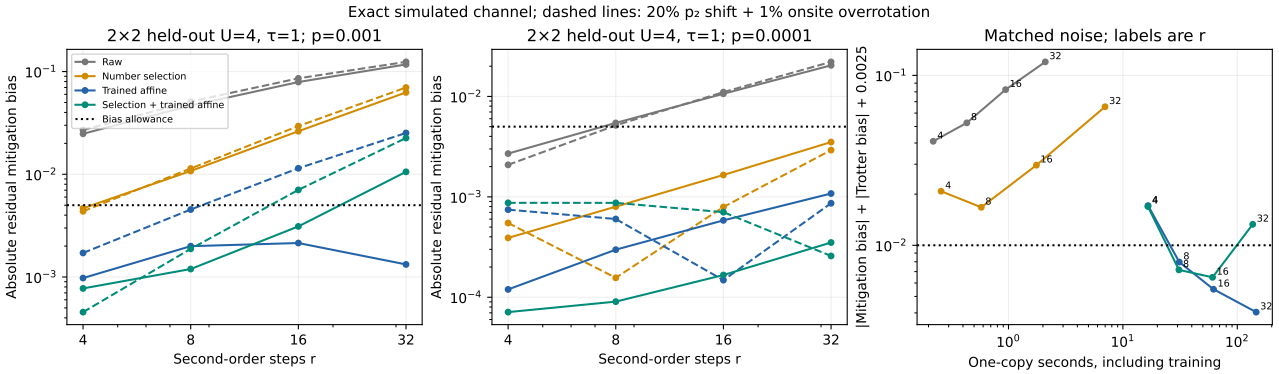


Figure 14: Held-out 2×2 estimator checks. The left panels show residual bias from the ideal circuit, separately from Trotter error. Dashed curves increase test-circuit two-qubit noise by 20% and onsite angles by 1%, without retraining. The right panel adds absolute mitigation and Trotter biases to the Gaussian statistical allowance and charges training time. Labels give the step count. The channel is simulated exactly; the shot plan is a Gaussian/delta-method approximation. No large-lattice or hardware accuracy guarantee follows.

5 Testing the assumptions behind the capacity curves

The capacity comparison depends on three implementation claims: that a physical estimator can afford its residual bias, that an angle-dependent STAR source retains favorable error and acceptance after correction and growth, and that FT arithmetic savings survive delivery and routing. A fourth question is whether a smaller physical patch can reproduce the chosen observable. We test each claim with a bounded calculation. These checks add evidence at their stated scale; they do not silently replace the full-campaign assumptions in Fig. 11.

5.1 NISQ: an estimator with measured residual bias

We simulate the full eight-qubit density matrix of a 2×2 plaquette circuit, including the native nearest-neighbor route, gate depolarization, idle noise, preparation, and readout. The noise convention is $p_2 = p$, $p_1 = p/10$, and idle error p per 100 ns. Spin-number selection retains outcomes with the correct up/down occupations. Following the training principle used in the experimental literature [10], we fit an affine correction to six noninteracting ($U = 0$) circuits and hold out all interacting tests ($U = 2, 4$). Training times differ from test times. Zero-angle interaction slots remain noisy, so the training circuits do not receive a cheaper noise history.

For selected mean D and acceptance A, the asymptotic variance per physical attempt is $D(1-D)/A$. If the affine estimator is $a + bD$, its test variance is multiplied by b^2. Each of the six training circuits receives 10^6 attempted shots in the resource model. With training uncertainty V_{train} propagated through both fitted coefficients, we choose

$$S = \left\lceil \frac{b^2 D(1-D)/A}{(\epsilon_{\text{stat}}/1.96)^2 - V_{\text{train}}} \right\rceil, \qquad \epsilon_{\text{stat}} = 0.0025, \tag{17}$$

when the denominator is positive. The fit uses population noisy training means to isolate systematic transfer error; an independent binomial training simulation checks the finite-shot variance approximation. The resource time charges the full training campaign to each test circuit.

At $p = 10^{-3}$, raw estimation fails the 0.005 bias allowance on all 24 matched-noise interacting tests. Number selection passes 8, affine training passes 10, and their combination passes 12. The latter falls

to 9 under the specified mismatch. At $p = 10^{-4}$, both trained estimators pass all 24 matched-noise bias tests. These are counts over $r = 4, 8, 16, 32$, $U = 2, 4$, and $\tau = 0.5, 1, 2$, not independent hardware trials.

The useful point is the coexistence of success and failure. For $U = 4$, $\tau = 1$, $r = 8$, and $p = 10^{-3}$, selection plus training leaves a mitigation bias of 0.00120, a Trotter bias of 0.00347, and a 0.0025 statistical allowance. The planned cost is 139,956 test attempts plus six million training attempts, or 30.9 s for one copy under the declared clocks. At $r = 32$, the same estimator leaves bias 0.01057 and fails the bias allocation. Refining the formula therefore does not necessarily improve the noisy answer. These measured outcomes constrain an estimator on this patch; fitting a size-independent mitigation exponent to them would require additional evidence.

5.2 STAR: accepted channels and growth components

We enumerate the projected error-angle mixture for the multirotation construction [4]. For odd injection distance d_i, let $k = (d_i + 1)/2$ and $\alpha = \arctan(|\tan\theta|^{1/k})$. With ideal syndrome projection, the no-fault acceptance is

$$P_{\mathrm{acc}}^{(0)} = \cos^{2k}\alpha + \sin^{2k}\alpha. \tag{18}$$

Independent rotation-generator Pauli faults have probability $p/15$ on weight-two factors and $p/30$ on the final weight-one factor. Enumerating the surviving branches gives weights w_j and angle errors δ_j, with coherence multiplier $c = \sum_j w_j e^{-2i\delta_j}$. This is an exact calculation for these faults and perfect projection; the other physical faults of an injection circuit are not included.

Ideal probabilistic coherent-error cancellation (PCEC) composes this channel with the conjugate mixture, giving $|c|^2$ and hence a Pauli channel. We also execute one level of noisy PCEC in the model: sampled inverse-angle corrections themselves use raw repeat-until-success (RUS) channels. Their angle-dependent preparation time, retries, and growth noise are charged. The resulting imaginary component is retained, rather than discarded by an assumed twirl. For a histogram with rotation multiplicities R_j, the separate ledger is

$$\chi = -2\sum_j R_j \log(\mathrm{Re}\, c_j), \qquad B_{\mathrm{coh}} \le \tfrac{1}{2}\left[\prod_j \left(1 + \frac{|\mathrm{Im}\, c_j|}{\mathrm{Re}\, c_j}\right)^{R_j} - 1\right]. \tag{19}$$

Here χ is the dephasing-component inverse-norm estimate, while B_{coh} bounds the coherent residue after that inverse; the product accounts for its non-CPTP norm. It is not a claim that the complete noisy gadget is Pauli. Workload retry caps reserve 1% of ϵ for omitted branches; the actual aggregate tail is recorded.

Detector rejection and high-distance growth remain explicit scenarios. We use $P_{\mathrm{acc}} = P_{\mathrm{proj}}(1 - c_d p)^{d_i^2+6d_i}$, with $c_d = 1, 4$, and independent growth dephasing

$$f(d_i, p) = \min\left\{0.25, f_9\left(\frac{p}{10^{-3}}\right)^5\left(\frac{d_i}{9}\right)^2\left(\frac{0.01}{p}\right)^{-(d_i-9)/2}\right\}, \quad f_9 \in \{0, 10^{-5}, 10^{-4}\}. \tag{20}$$

There is no growth term when $d_i = d_{\mathrm{data}} = 25$. The f_9 values are anchored at $p = 10^{-3}$; their assumed extension scales as $p^{(d_i+1)/2}$ at fixed distance. These coefficients are not transplanted calibrations from Chung et al., whose reported noise model differs [7]. Switching chooses the original-injection error reference when it is smaller after growth; its error and high-acceptance service remain declared assumptions. The study therefore tests a mechanism and its sensitivities, not the feasibility of a complete optimized source.

A separate circuit-level experiment tests direct rotated-patch growth from $d_i = 3, 5$ to $d_f = 5, 7$, with no-growth controls. The starting patch is ideal; fresh right and bottom strips are prepared in Z

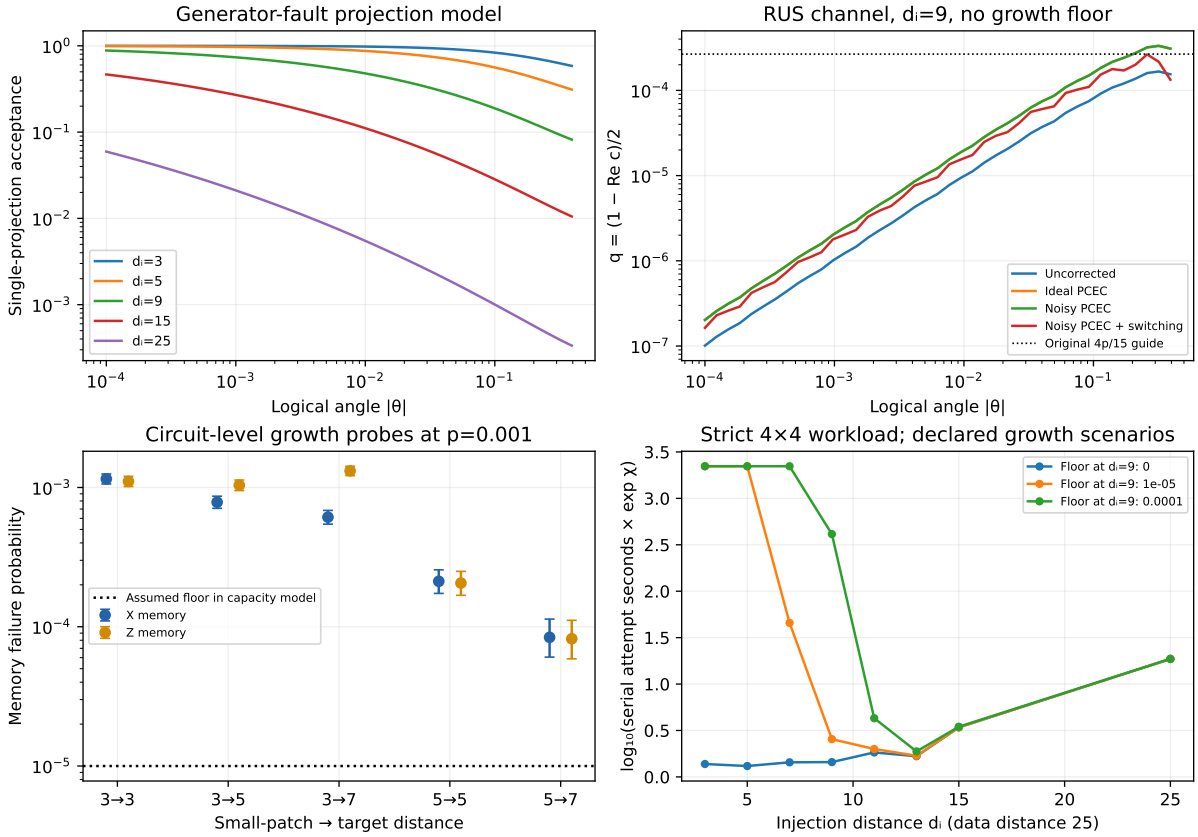


Figure 15: STAR mechanism and growth checks. Upper panels use the specified rotation-generator fault submodel; $q = (1 - \mathrm{Re}\, c)/2$ alone does not describe its coherent component. Lower left: decoded X/Z memory error for ideal small-patch preparation followed by noisy growth, stabilization, and readout, with fixed-sample 95% binomial intervals. The dotted floor is a capacity-model assumption, not a fitted growth curve. Lower right: serial service times multiplied by the dephasing sampling factor for a fixed strict-precision 4×4 workload, under declared growth scenarios. A complete injected-state error model is still required to replace the STAR capacity assumptions.

and X respectively, then the target patch is stabilized for d_f rounds. Stim supplies a nearest-neighbor extraction schedule; transition detectors are constructed from exact stabilizer flows and later detectors compare adjacent rounds. PyMatching decodes the circuit. Gate, reset, measurement, and idle errors are included, with the 240 ns cycle explicitly charged. Every one of 20 cases uses 500,000 shots, including both X/Z tests and both physical error rates. Noiseless and deliberately inserted logical-flip controls check that the detector and logical records are meaningful. These memory components include final readout and exclude noisy initial injection; they cannot establish the distance-25 arbitrary-angle growth floor.

A larger fixed-shot campaign extends the same protocol to initial distances $d_i = 5, 9, 13$ and final distances $d_f = 13, 25$ at $p = 10^{-3}, 10^{-4}$ in both bases. All 24 cases complete one million shots, following a two-case PBS topology smoke. Both jobs finish with zero exit status, and the case census retains seeds and circuit hashes. At $p = 10^{-3}$, $5 \to 25$ produces 51 X-basis and 419 Z-basis failures; $5 \to 13$ produces 60 and 175. Every other case has zero observed failures. A zero count has simultaneous one-sided upper bound 6.174×10^{-6} per component, using a 5% family error budget over all 24 intervals. If the two basis tests describe a common Pauli channel, their union bounds its total error by 1.235×10^{-5} for zero-failure pairs. Thus these data constrain growth at the data-code distance but do not yet certify

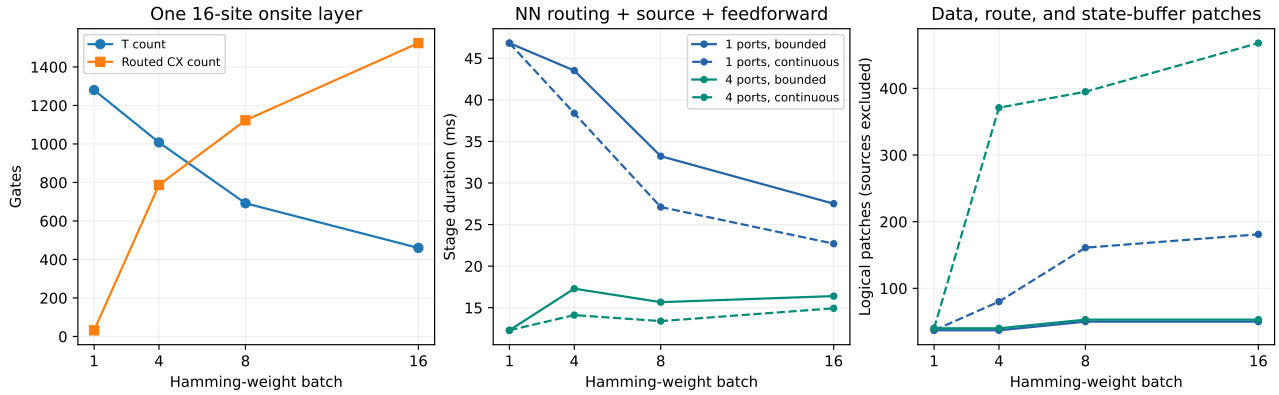


Figure 16: One 4×4 onsite layer at $\theta = 1/36$. Hamming-weight batching reduces T count but increases arithmetic and routed CX count. One-state buffers and continuous production distinguish timing from storage assumptions. The patch count includes packed data/routing and state buffers; source and surgery footprints are excluded. This constructive schedule neither optimizes routing nor establishes full-campaign physical area.

a 10^{-5} total floor. They also show that the initial patch distance matters even when the final patch is large. Initial injection faults remain excluded, and we do not subtract the no-growth control from a measured campaign error.

5.3 FT: arithmetic savings versus a routed schedule

An explicit onsite layer on 16 sites tests the repeated-angle count model. Parity CNOTs turn $e^{-i\theta Z_\uparrow Z_\downarrow}$ into equal R_z rotations. The Hamming-weight compressor uses $b - \mathrm{popcount}(b)$ clean carries per batch [13]. Each temporary AND uses four T states and is exact on a clean target; X measurement and a conditional CZ uncompute it without extra T gates. Independent tests check every binary input through $b = 16$ and arbitrary superpositions in every measurement branch through $b = 4$. CUDA-Q Logical compiles the complete arithmetic and feedforward program. GridSynth counts at precision 10^{-8} agree with the explicit arithmetic plus the per-angle synthesis inventory.

We route every two-patch operation to adjacent square-grid patches and schedule dependencies, source availability, and feedforward. Declared macros take three d-round clocks per CX/CZ, one per H/S or T consumption, two for X measurement/reset, and an additional feedforward clock. A source port emits one state per 150 physical rounds. With $d = 25$ and 240 ns rounds, the bounded policy holds one state per port and pauses production while occupied; continuous production instead reports its storage demand. The shared ports constrain availability; magic-state transport paths are not geometrically routed. These are logical macros, not a decoded physical surgery layout.

Batch 16 reduces the stage from 1280 to 460 T states, but raises routed CX count from 32 to 1524. With one source port and bounded storage, the stage falls from 46.824 to 27.510 ms. With four ports it instead rises from 12.282 to 16.398 ms. Continuous production reduces waiting but can require hundreds of buffer patches. Thus the aggregate Hamming-weight resource improvement is supported at the T-count level; its runtime and area improvement require a better joint routing, arithmetic, and delivery implementation than is assumed by a count-only comparison.

We then replace abstract delivery by a conservative geometric construction. A magic state enters at a perimeter output port, moves by adjacent SWAPs through the patch array, and is consumed next to the target. Reversing the SWAPs restores every displaced data qubit. Tests verify the complete channel on entangled data in both injection outcomes. Each SWAP costs three adjacent CX operations,

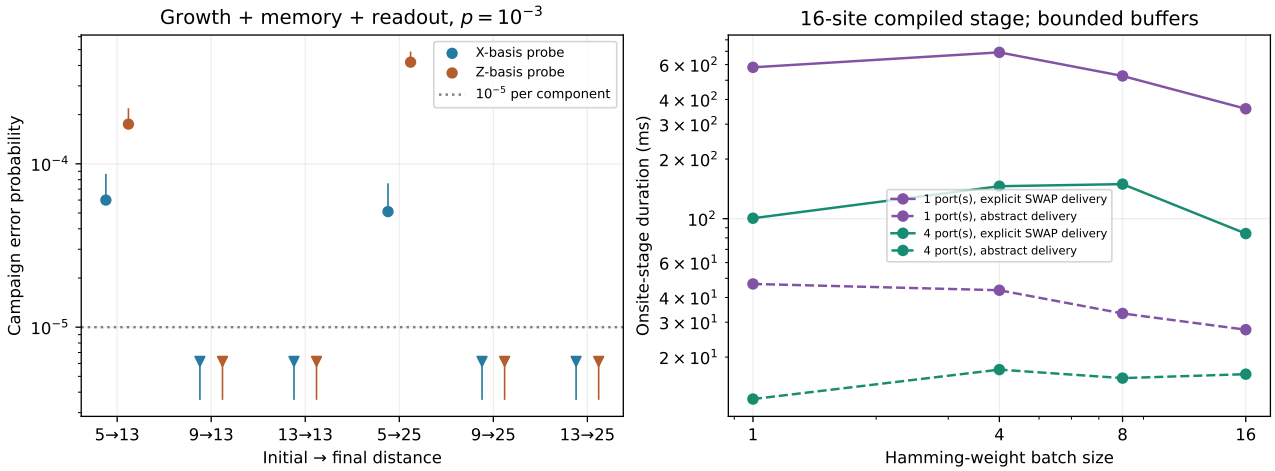


Figure 17: Larger growth components and geometric FT delivery. Left: one million shots per basis at $p = 10^{-3}$. Circles are measured rates, with segments reaching simultaneous upper bounds; downward triangles mark upper bounds where no failures were observed. The experiments include target memory and readout but start from an ideal encoded state. Right: the same compiled onsite stage with abstract delivery (dashed) or explicit SWAP transport and restoration (solid). Port count, state supply, and data routing are held fixed within each pair. The geometric construction is deliberately conservative and does not establish an optimal surgery layout.

and the injection takes five logical clocks. The schedule reserves the traversed patches and their surgery mediators until return. Embedding patch centers at pitch two and retaining the perimeter gives 225 tiles for batches one and four, or 289 for batches eight and sixteen. These footprints include gaps and corners; the reference source area is charged separately, with one bounded output buffer per port.

With one port, explicit delivery changes the batch-one and batch-16 stage times to 581.118 and 358.890 ms. Four ports reduce these to 100.518 and 84.102 ms. Batches four and eight can be slower despite using fewer T states. The ranking is therefore a property of the arithmetic, placement, and transport together. We retain this isolated stage as a separate layout scenario rather than extrapolating its transport penalty to every operation in the capacity plot.

5.4 Locality: the observable and its Trotter error differ

We compare exact doublon values on nested patches with the same Néel pattern, $U = 4$, and observed coordinate $(1, 1)$. The smaller patches are recomputed; completed 4×3 and 4×4 exact runs supply finite references. At $\tau = 1$, $|D_{3\times3} - D_{4\times4}| = 0.04015$ and $|D_{4\times3} - D_{4\times4}| = 0.01488$. Every tested proper subpatch of 4×4 exceeds the 0.0025 spatial allowance at $\tau = 1, 2$.

The similar plaquette Trotter coefficients near $\tau = 1$ describe the error of the algorithm relative to each patch's own exact dynamics. They do not establish convergence of the dynamics themselves. The new measurements reject the tested aggressive truncations, while leaving a properly centered larger-patch or independently bounded locality study as a distinct requirement. At fixed time, a validated local approximation would also change the task metric: once its patch fits, maximum full-lattice size no longer measures the cost of that one observable.

A state-independent certificate supplies a complementary sufficient condition. We center the observable, $O = D - I/2$, so $\|O\| = 1/2$, and remove onsite interactions by a product interaction picture. A two-spin hopping bond has norm $2J$. The commutator recursion [17] and Duhamel's formula give

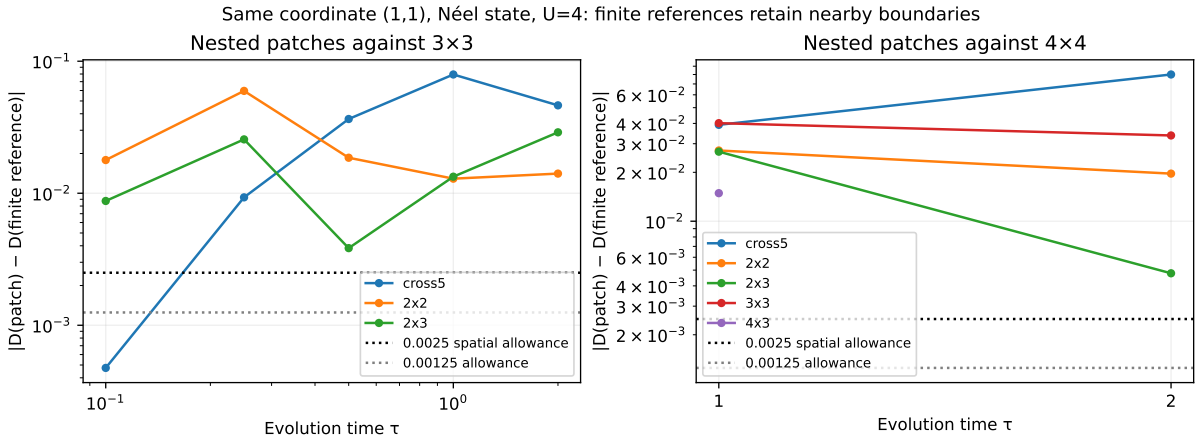


Figure 18: Direct nested-patch differences with a fixed observed coordinate. Each curve compares to the named finite reference. Neither reference removes the nearby outer boundary, so the data do not certify an infinite lattice or a saturation length. Agreement of Trotter-error coefficients must not be used as evidence that these observable differences are small.

the deliberately loose bound

$$\delta_{\text{spatial}}(R,\tau) \le \min\left\{1, \frac{2}{7}\sum_{n\ge R+1}\frac{(28|\tau|)^n}{n!}\right\}. \tag{21}$$

Here the retained patch contains the graph ball of radius R around the observed site, and its initial reduced state agrees with the full system. There are at most four first bonds and seven bonds touching each subsequent bond. The first omitted boundary crossing needs at least $R+1$ bonds; Appendix D details the counting. The geometric tail majorant first meets a 0.0025 allowance at $R = 78$ for $\tau = 1$, corresponding to an enclosing 157×157 square. This sufficient radius is not an estimate of the physical light cone. The bound is too loose to certify the proposed six- or eight-site-wide patches, while the direct finite-reference checks reject the smallest truncations. A tight bulk truncation estimate remains separate scientific work.

5.5 Do the measured costs collapse onto exposure?

The exposure laws organize the accounting, but a collapse requires an independently obtained ordinate. We use exact noisy eight-qubit expectations, held-out affine fits, and enumerated STAR channels rather than plotting an assumed variance law against its own exponent. For each NISQ circuit, χ_{PEC} sums the gate, idle, and SPAM inverse-norm contributions. The measured variance multiplier divides the estimator variance per attempt by $D_{\text{ideal}}(1 - D_{\text{ideal}})$; acceptance and fitted slope are included. Training uncertainty and all six million training shots per fit remain in the campaign ledger, although the ordinate isolates inference variance.

The trained estimators have much smaller variance than the PEC reference, but some fail the held-out bias requirement at the same exposure. These data do not support a universal variance-and-bias collapse onto one scalar. Likewise, STAR's normalized error depends on the correction channel and injection distance. The useful outcome is a test of the law's domain: exposure alone is insufficient to certify an estimator or a full injection protocol.

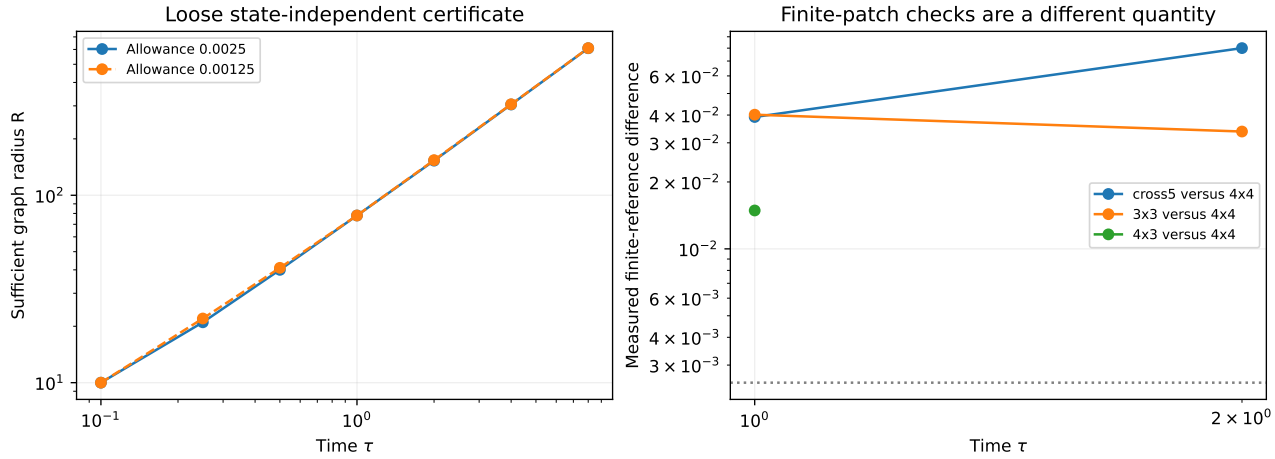


Figure 19: Sufficient locality radius from Eq. (21) (left) and measured finite-reference differences (right). These are different quantities: a loose uniform bound cannot locate the physical light cone, and agreement with one finite reference cannot bound its outer boundary.

6 Reconciliation of the workshop calculations

We evaluate the two supplied workshop implementations at 4×4, $\tau = 1, 2$, and 0.01 absolute error, with common elementary clocks and a common Gaussian doublon shot budget. The statistical allowance is 0.004 at $D_\star = 0.1$, giving 21,609 ideal shots. Taking 25% of the error budget for the larger of our two planning coefficients gives 11 and 23 steps. This step convention isolates resource differences; it is not an accuracy certificate for the different observables and unspecified circuits underlying the supplied models. We remove the IQUS physical-cone discount and explicitly fix its signal scale. The original implementations remain untouched, and their imported Python sources are preserved alongside the comparison data.

Matching these settings does not make the resource inventories identical. At $\tau = 1, r = 11$, the two-qubit gate counts are approximately 4,750, 3,520, and 2,640 for our native proxy, Martin, and IQUS, respectively. The corresponding STAR rotation counts are 2,443, 2,288, and 880. Their STAR shot times are approximately 50.3, 1.86, and 0.686 ms, reflecting different rotation parallelism and retry schedules even after the syndrome clock is matched. Native factory and logical-error conventions also remain different. These outputs diagnose assumptions rather than rank three fully specified machines.

An additional comparison applies every injection law to the identical ledger $R = 2{,}443$, $\Theta = 64$ at $p = 10^{-3}$. Martin's native attenuation convention gives $\chi = 0.9772$, the IQUS count convention gives 2.6042, the exact original-injection inverse gives 2.6066, and the angle-plus-floor law gives 0.4817. Thus even the phrase "count law" does not identify a common estimator convention. The differences do not collapse onto precision, evolution time, syndrome clock, and one injection switch alone: compilation, delivery, and the estimator must also be named. This shared-ledger calculation is an algebra check, not an empirical collapse test or independent evidence for a mitigation law.

7 Back-of-the-envelope+ discussion example

We use simple analytical laws to expose the architectural tradeoffs, with $p = 10^{-3}$, 10 ns gates, 100 ns measurement/reset, 240 ns syndrome cycles, $U/J = 4$, and a one-week budget B. The original reference below emphasizes $\epsilon = 0.001$; both precisions remain explicit in the current comparison of Section 4. We compare NISQ, STAR, FTQC, and the hybrid using a common total-error target. Each

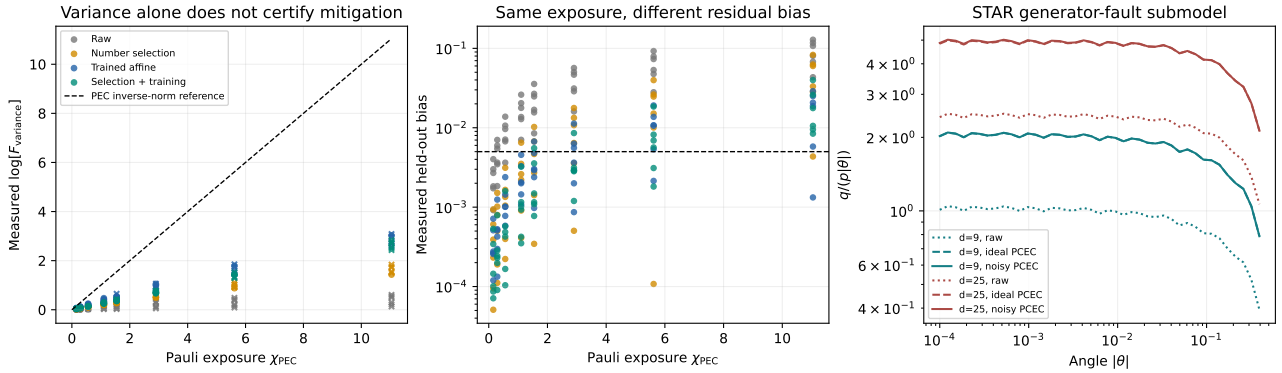


Figure 20: Independent exposure diagnostics. Left and center: matched-noise held-out tests at $p = 10^{-3}, 10^{-4}$, with crosses marking failure of the 0.005 bias allowance in the left panel. Small variance does not imply small bias. The dashed diagonal is the separate whole-circuit PEC inverse-norm reference. Right: $q/(p|\theta|)$ from the exact generator-fault submodel at $p = 10^{-3}$, separating raw, ideal-PCEC, and noisy-PCEC channels. Generic extraction faults and growth are absent from these channel curves.

arm can choose its error allocation, step count, and resource-state supply within the declared search. This removes a fixed allocation that previously forced unnecessary cleanup at the strict 4×4 FT point. The current finite-patch capacity sensitivity is shown in Figure 27, and the directly checked 4×4 allocations in Table 3. We first develop the common model using the original 3×3 reference, retaining its longer-time and $\tau = L/4$ scenarios for comparison.

Ordering-specific time dependence. We extend exact fixed-sector calculations on 2×2 and 3×3 patches through $\tau = 8$, and check the two product formulas separately. The termwise evolution retains the ordering of the compiled Pauli circuit; the plaquette evolution retains its onsite/pink/gold splitting. Independent small-system comparisons check the factorized sector implementation against the qubit circuits and the sparse Hamiltonian evolution.

For each ordering $\mathcal{A}$ and measured time, we increase r until the last two estimates of $|D_r - D_{\text{exact}}|r^2$ agree within 5%. The observed r^{-2} dependence survives, but its coefficient is neither a single power of time nor independent of the ordering. On 3×3, the plaquette and termwise coefficients are respectively 0.218 and 0.0767 at $\tau = 1$, and 7.67 and 23.74 at $\tau = 8$. Figure 21 shows the full time dependence and patch sensitivity. At $\tau = 2$, the 4×3 plaquette and termwise coefficients are 1.136 and 0.702, compared with 0.926 and 0.381 on 3×3. At $\tau = 4$ they are 4.100 and 2.728, compared with 5.270 and 1.957. These differences preclude a claim of bulk convergence.

At the smaller step counts admitted by the optimizer, higher-order terms can matter. We therefore also measure $r = \lceil 8\tau \rceil$ and twice that value on 3×3. The coefficient used below is the largest sampled value across these finite-step checks and the asymptotic pair. This corrects an underestimate caught by direct validation at loose precision.

For the planning curves we use

$$C_{\mathcal{A}}^{\text{plan}}(\tau) = 1.1 \max_{s \le \tau,\ s \in \mathcal{T}} C_{\mathcal{A}}^{3\times 3}(s), \qquad \epsilon_T(r,\tau) = \frac{C_{\mathcal{A}}^{\text{plan}}(\tau)}{r^2}, \tag{22}$$

where $\mathcal{T}$ is the measured time grid. The running maximum avoids relying on isolated small-patch cancellations; it is a planning choice for the final-time observable, not a change to a whole-curve accuracy target. The 10% margin is a declared sensitivity, not a finite-size error bound. We retain

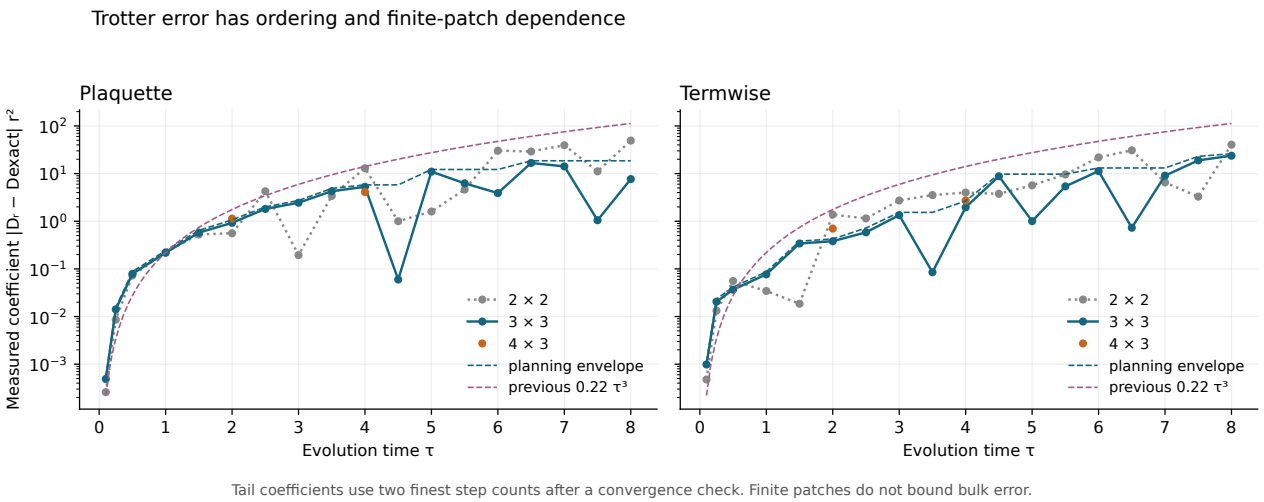


Figure 21: Local Trotter-error coefficients depend on time, ordering, and patch. Each point uses the finer pair after a step-convergence check. The dashed blue curve is the planning envelope in Eq. (22); the previous $0.22\tau^3$ prescription is shown for comparison. The patch differences are not a bulk error bound.

pointwise-coefficient costs separately. No time power is fitted, and we do not extrapolate beyond $\tau = 8$. The 2×2 resource point uses its own measured coefficients; other sizes transfer the 3×3 reference. The $\tau = L/4$ panel therefore stops at $L = 32$. Transfer to larger lattices remains an assumption.

Joint error allocation. For each candidate we impose

$$\epsilon_{\text{stat}} = \epsilon - \epsilon_T - \epsilon_L - \epsilon_{\text{syn}} - \Gamma N_T q_T - \epsilon_{\text{res}} > 0. \tag{23}$$

Unused components vanish: termwise STAR has no synthesized rotations or supplied T states, and the hybrid synthesizes no variable rotations. The logical reserve satisfies $\Gamma V_L p_L(d) \le \epsilon_L$. Mitigated arms reserve $\epsilon_{\text{res}} = 0.1\epsilon$ for residual estimator bias, including channel mismatch and unmodeled SPAM. Its affordability is assumed. FT has no mitigation reserve; its modeled faults are charged to the logical and magic-state terms.

We search Trotter allowances, logical reserves, synthesis tolerances, and integer-step neighbors, with $r \ge \lceil 8\tau \rceil$ as a small-step guard. FT and hybrid independently compare raw cultivated states with one cleanup stage and optimize 1–16 sources. Both use the same source menu. The result is the best candidate in the declared search, not a global optimum over protocols or source-quality/throughput tradeoffs. The resulting per-arm allocations are recorded with each resource point.

For effective mitigation and encoded planning curves,

$$M_0 = \left\lceil \frac{1.96^2 D_*(1 - D_*)}{\epsilon_{\text{stat}}^2} \right\rceil, \qquad Q_{\text{week}} = q \max\left(1, \left\lceil \frac{M_0 F t_{\text{shot}}}{B} \right\rceil\right), \qquad D_* = 0.1. \tag{24}$$

The PEC references retain the sufficient Hoeffding count from Eq. (33). Distance follows the common surface-code fit, with worst-case $\Gamma = \sqrt{F}$ amplification for mitigated logical and magic faults. The central delivery model has four effective parallel lanes; these are a throughput assumption, not a compiled dependency schedule.

The per-arm laws. NISQ uses the Willow circuit-count anchor [10],

$$G_2 = 4372 \frac{r}{3} \left(\frac{L}{6}\right)^3, \qquad D_{\text{circ}} = 294 \frac{r}{3} \frac{L}{6}, \qquad \log F_{\text{eff}} = \log(\eta_*) \frac{p}{10^{-3}} \frac{G_2}{4372}. \tag{25}$$

The central $\eta_* = 100$, with 10–1000 sensitivity, is an assumed sampling overhead, not a fit to experimental accuracy. Its bias reserve also remains uncalibrated. Two explicit PEC comparisons use two-qubit inverse-channel exposure, $\log F \simeq 4pG_2$, or the saved compiled full-noise exposure with bulk terms proportional to r and separate SPAM. The NISQ proxy transfers the plaquette error coefficient without asserting a match to the experimental ordering.

Termwise STAR uses

$$\Theta = (5L^2 - 4L)\tau, \qquad R = [16L(L-1) + 2L^2 - 2]r + 1, \qquad \log F_{\mathrm{STAR}} = 4\alpha p\Theta + 4fR, \tag{26}$$

with $\alpha = 1.5$, $f = 10^{-5}$, and 18 code clocks per completed rotation [12, 4]. The scenario range varies f from zero to 10^{-4} and lanes from eight to one. Refinement leaves the angle term unchanged while increasing the floor term and execution time. FT uses plaquette counts, logarithmic synthesis, and $F = 1$. The hybrid uses the same plaquette workload, injecting variable rotations and supplying fixed T states. It therefore has exponent $4\alpha p\Theta_{\mathrm{dyn}} + 4fR_{\mathrm{dyn}}$.

FT and hybrid share the rounded cultivation reference: 1024 source qubits, 100 syndrome cycles per state plus $2d$ handoff cycles, and raw state error 2×10^{-9} [18]. The optional 15-to-1 cleanup changes source error, footprint, and service together. Supply overlaps consumption; the reference charges nine code clocks per T and varies source service time by 0.5–4. These source models remain conditional implementations. Termwise STAR already avoids fixed T gates, so a hybrid advantage is algorithm-dependent.

Using the 3×3 planning reference at $\tau = 1$, the central areas are

Lattice	ϵ	NISQ effective	STAR	Hybrid	FTQC
4×4	.001	36	71442	73348	1.43×10^6
4×4	.01	36	46818	48420	61974
6×6	.001	3.64×10^{10}	298356	180798	1.16×10^7
6×6	.01	81	101802	103548	126268

For strict 4×4 FT, the selected $r = 31$ candidate allocates about 33.6% to statistics, 25.3% to Trotter error, 36.2% to magic-state error, 2% to logical faults, and 3% to synthesis. Cleanup is no longer needed, reducing the previous 5.66 million-qubit estimate by about a factor of four. The strict 6×6 FT candidate still chooses cleanup. The hybrid improves on termwise STAR at strict 6×6, but not at 4×4. NISQ's effective estimate remains conditional on the specified sampling multiplier and affordable residual bias; its PEC alternatives are shown alongside it.

We separate the changes with an ablation that first optimizes the budget under the old cubic-time coefficient and then replaces that coefficient. A finer allocation grid improves the selected $\tau = 1$ costs by at most 18%. Direct checks at the optimized step counts fit the planning envelope on 3×3. At strict 4×4, $\tau = 8$, the optimized STAR floor exposure is 2.28, below its angle exposure 3.07; the earlier floor-dominated example depended on the assumed cubic time law. Neither validates the transfer from finite patches to large lattices. The earlier four-step compiled reference and fixed-allocation discussion results are retained as historical comparisons. The remainder develops their resource laws and implementation evidence.

8 Delivery rate and protected area

Faster consumption helps only when its extra area and state supply are included. We reoptimize the error budgets, source count, and replicas for three data-block designs from Litinski [19]. For n stored logical qubits their tile counts and worst-case T-consumption times are

$$(A, c_T)_{\mathrm{compact}} = (1.5n + 3, 9), \quad (A, c_T)_{\mathrm{intermediate}} = (2n + 4, 5), \quad (A, c_T)_{\mathrm{fast}} = (2n + \sqrt{8n} + 1, 1). \tag{27}$$

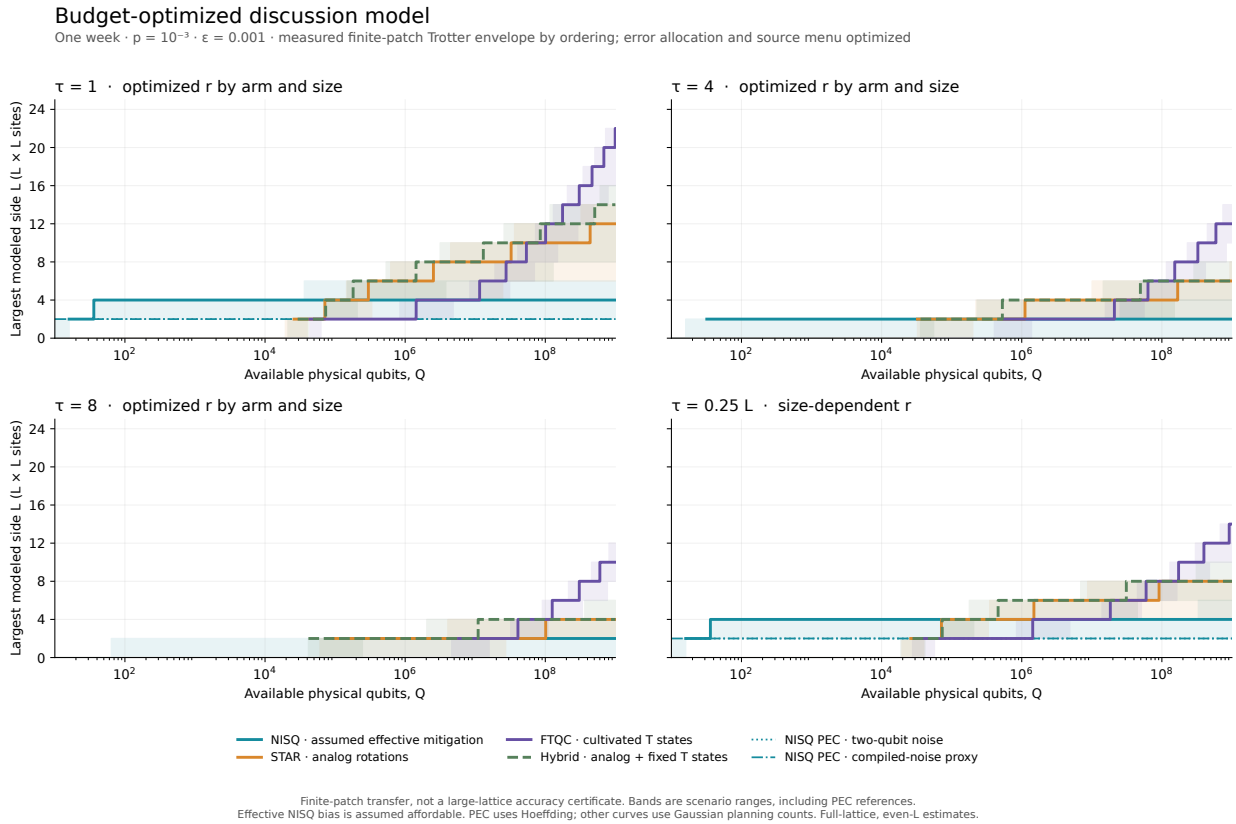


Figure 22: Budget-optimized discussion comparison at $\epsilon = 0.001$. Each arm selects its error allocation and step count using the measured finite-patch planning envelope. Lines are central assumptions; bands span delivery and mitigation scenarios, including NISQ PEC references. They do not represent uncertainty in bulk Trotter error. The fourth panel follows $\tau = L/4$. Full lattices and even L are modeled. Effective NISQ mitigation is an assumed estimator scenario, not a demonstrated accuracy result.

Here c_T is in code clocks, each containing d syndrome cycles. We use $n = 2L^2 + 6$, round the tile count upward, and retain six additional tiles per delivery or injection lane. The fast-block expression is a reference area formula, not a placed floorplan. Its relative overhead at small L is larger than the asymptotic $4/3$ ratio.

The earlier discussion model already assumes four effective lanes and divides its nine-clock T cost by four. It is therefore not the paper's serial compact block. For the whole-block comparison we do not divide the published T rate by the analog-lane count. We retain a separate, explicitly uncompiled scenario with four fast T ports, charging the fast data area and four lane allowances. Whether those ports can execute independent Pauli products at that rate requires a dependency and routing construction.

STAR retains eighteen clocks per analog rotation. Its effective parallelism is varied over $b_A \in \{1, 2, 4, 8, 16\}$, with area charged for every lane. The hybrid varies analog lanes and T consumption independently. For m sources and b_T T ports, the competing stages are

$$t_{\text{supply}} = \lceil N_T/m \rceil \, s(d) t_{\text{cyc}}, \tag{28}$$

$$t_{\text{consume}} = d \left[\frac{c_T N_T}{b_T} + \frac{18 R_{\text{analog}} + C_{\text{Cliff}}}{b_A} \right] t_{\text{cyc}}. \tag{29}$$

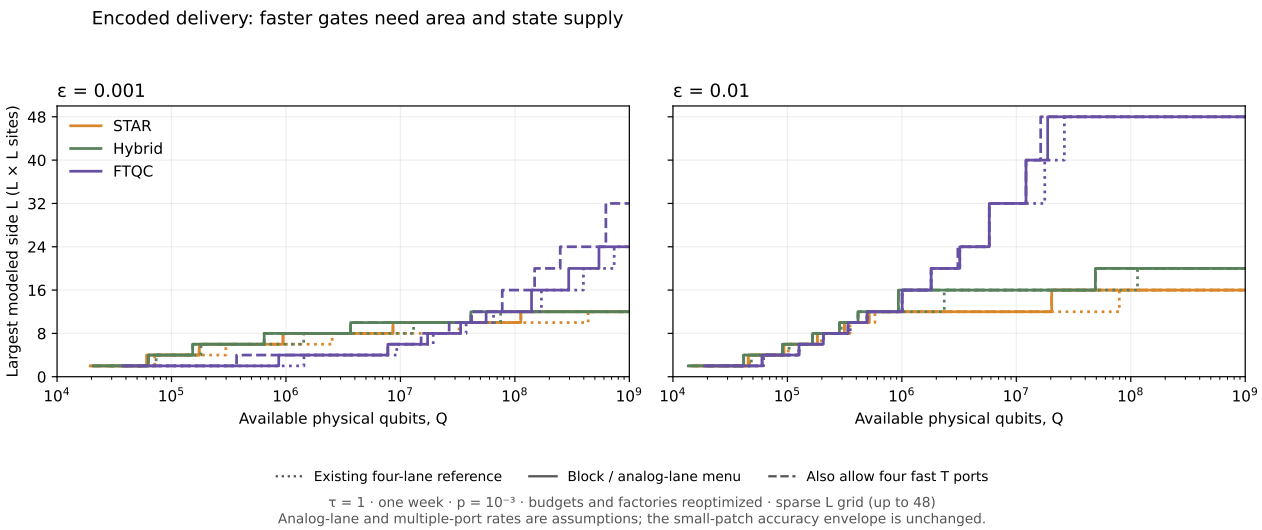


Figure 23: Encoded capacity at $\tau = 1$ after reoptimizing budgets and state supply. The dotted reference retains four effective lanes. The solid menu adds whole-block designs and analog-lane choices; the dashed menu also allows four fast T ports. Menus are nested and share the expanded source choices. Parallel rates are conditional assumptions. Staircases use only the sampled sizes, with $L = 48$ the sweep endpoint.

The shot duration uses their maximum, plus initialization and feedforward. The whole-block T rate sets $b_T = 1$. Both FT and hybrid receive the same source menu, extended through 1024 sources so faster consumers can be supplied; selected optima must lie below that endpoint. The menu search is discrete, and analog parallelism remains an assumption rather than a compiled schedule.

At strict 4×4, $\tau = 1$, the single fast block reduces FT's 1.43 million-qubit reference to 0.86 million. Four fast ports reduce it to 0.37 million, where state supply limits further speedup. Thus the reference does not receive a universal factor-of-nine improvement. STAR and hybrid select two analog lanes at this point, requiring about 61,000 and 63,000 qubits. At loose precision one replica often fits the week; then extra throughput can increase total area. Figures 23 and 24 separate these effects. These sensitivities retain the same 3×3 planning coefficients, so they isolate delivery from finite-size calibration.

9 Direct step checks and finite-patch time dependence

We compute the exact reference evolution in the $N_\uparrow = N_\downarrow = 8$ sector of the 4×4 OBC lattice. Its $\binom{16}{8}^2 = 165{,}636{,}900$ amplitudes fit a matrix-free calculation with four complex state vectors, using about 10 GB of memory. At $\tau = 1$, the central doublon at coordinate $(2, 2)$ is 0.1452313819. The sampling convention remains $D_* = 0.1$ and the absolute targets remain 0.001 and 0.01; we do not silently redefine the requested precision using the newly calculated signal.

The plaquette coefficient $|D_r - D_{\text{exact}}|r^2$ changes from 0.262431 at $r = 16$ to 0.261828 at $r = 32$, a 0.23% difference. It is about 20% above the 3×3 value. The termwise coefficient changes from 0.188386 to 0.187732, a 0.35% difference, and is about 2.45 times its 3×3 value. The 4×3 coefficients, 0.2453 and 0.1392, lie between the square-patch results, as Figure 25 shows. The earlier 10% planning margin therefore does not cover even this $\tau = 1$ transfer. Exact finite-lattice evolution resolves that specific uncertainty; it does not establish an infinite-lattice coefficient.

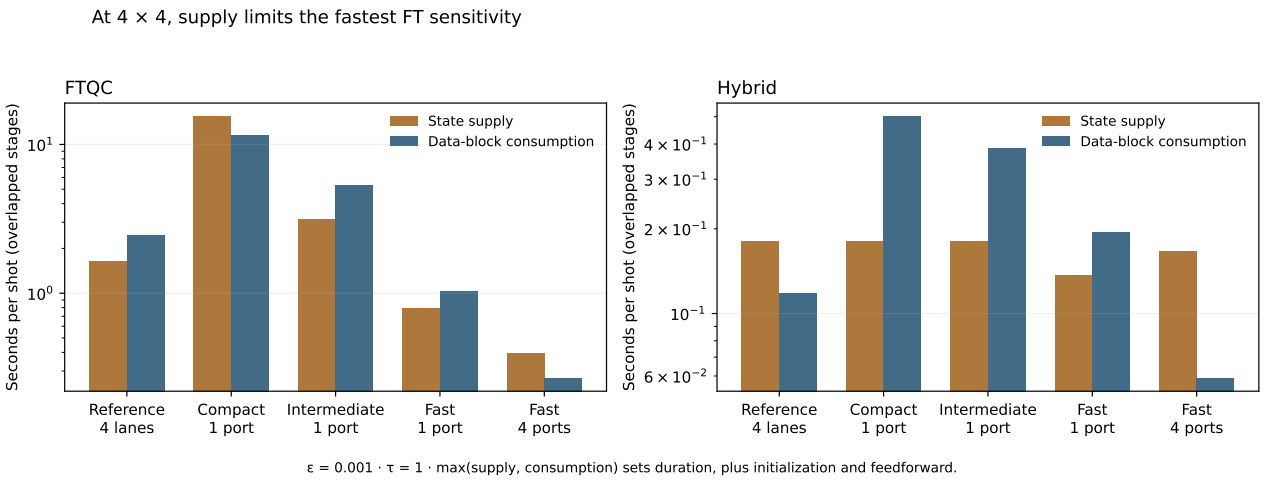


Figure 24: Supply and consumption durations at strict 4×4, $\tau = 1$, with each layout's budgets and source count reoptimized. These stages overlap: their maximum enters shot time. The fastest FT sensitivity becomes supply-limited, while the hybrid also pays for its analog rotations.

Design	Saved r	Saved Q	Measured-error budget Q
NISQ effective	21	72	72
STAR reference	33	71,442	71,442
Hybrid reference	41	73,348	73,348
FT reference	33	1,875,882	1,655,190
FT single fast block	29	1,155,490	988,112
FT four fast ports	33	385,052	385,052

Table 3: Direct 4×4 checks at $\tau = 1$, $\epsilon = 0.001$. Both columns use the same step count, hardware, and source menu. The last column reallocates the smaller measured Trotter error before minimizing resources. Effective NISQ retains its assumed mitigation bias and variance; four fast ports retain the uncompiled throughput assumption.

Validate before reallocating. The initial 4×4 resource sensitivity interpolated between measured steps using C/r^2, with a 10% margin. We now directly evaluate every selected step count for the headline reference designs: effective and two PEC NISQ models, reference STAR and hybrid, and reference, single-fast-block, and four-fast-port FT. The two precision targets give 16 allocations sharing nine circuits. All satisfy

$$|D_r - D_{\text{exact}}| + 10^{-10} \leq \epsilon f_T^{\text{saved}}, \tag{30}$$

using their *original* Trotter fraction f_T^{saved}. The measured errors use 89.3%–90.9% of these allowances. No change to statistics or another error component is needed to pass these checks.

We then separately reoptimize with the measured Trotter error and the same 10^{-10} numerical allowance. Table 3 shows the strict estimates at each originally selected r. Reference FT falls from 1.88 to 1.66 million qubits, and the single fast block from 1.16 to 0.99 million, because the smaller charged Trotter error leaves more allowance for sampling. STAR and hybrid remain single-copy calculations. The saved Trotter allowances were already adequate; these reductions come from using their remaining allowance. Searching all measured steps gives the same headline areas, sometimes at a different r.

Checking the selected NISQ count $r = 21$ removes the earlier artificial restriction to $r = 8, 16, 32,$

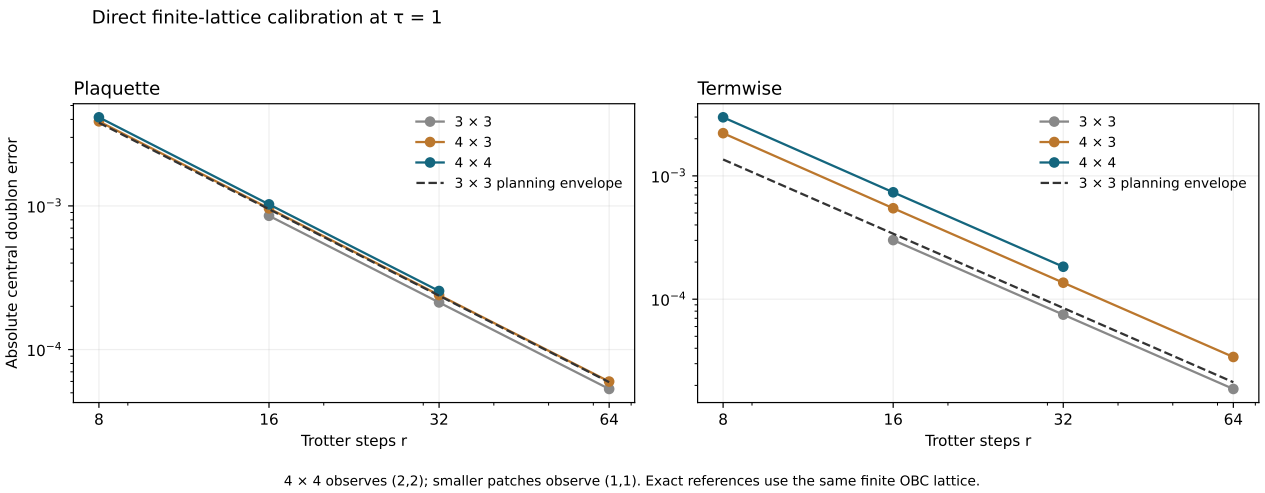


Figure 25: Direct Trotter errors at $\tau = 1$ on three finite patches, with exact references on each same lattice. Both orderings exhibit r^{-2} convergence, with coefficients that depend on the patch. The dashed line is the previous 3×3 planning envelope, including its 10% margin. The 4×4 observable is at $(2, 2)$; smaller patches use $(1, 1)$. This comparison calibrates the finite 4×4 example and does not certify large-L or long-time transfer.

which had inflated its measured-menu cost to 2196 qubits. This validates the Trotter component only: plaquette accuracy is still transferred to the experimental gate-count proxy, and mitigation bias remains unmeasured. At loose precision the two-qubit PEC scenario also fits one 36-qubit copy at $r = 8$, whereas its strict estimate exceeds 10^{12} qubits. The full-noise PEC proxy is much more costly. These differences remain visible in the saved tables and should not be read as an architecture-independent NISQ limit.

Longer time and the capacity sensitivity. We extend the same exact 4×4 reference and both product formulas to $\tau = 2, 4$. At each time, the initial counts are $8\tau, 16\tau, 32\tau$; further doubling is required if the two finest coefficients disagree by more than 5%. Figure 26 compares the converged tails with the smaller patches. The observed site remains $(2, 2)$ and each product formula is compared with exact evolution on that same lattice. The converged-pair coefficients on 4×4 are 1.096 and 0.775 for plaquette and termwise evolution at $\tau = 2$, and 4.193 and 3.029 at $\tau = 4$. The finest pairs agree within 0.36% in all four cases. The $\tau = 4$ plaquette value is close to 4×3 and below 3×3; the termwise value exceeds both smaller patches. A single multiplicative finite-size correction would miss this dependence on ordering and time.

To isolate this physics input, we keep the reference delivery assumptions and the expanded common source menu fixed and reoptimize the resource budgets under two coefficient policies. The first is the earlier 3×3 envelope. The second retains that envelope and adds all measured 4×3 and 4×4 coefficients through the requested time, including finite-step corrections, with the same 10% margin. The $L = 2$ point keeps its own calibration in both cases. Figure 27 reports the resulting capacity sensitivity. It does not replace a finite-patch assumption with a bulk error bound: the new curves still transfer coefficients to $L > 4$, and their optimized step counts have not all been checked directly. No 4×4 coefficient is inferred at $\tau = 8$. The additional patches move several discrete thresholds while retaining the strong precision dependence in this reference model. For example, at $\tau = 2$, $Q = 10^6$, and strict precision, effective NISQ falls from $L = 4$ to $L = 2$, whereas reference STAR and hybrid remain at $L = 4$. This is a sensitivity to the assumed mitigation and finite-patch models; it is not a general NISQ limit.

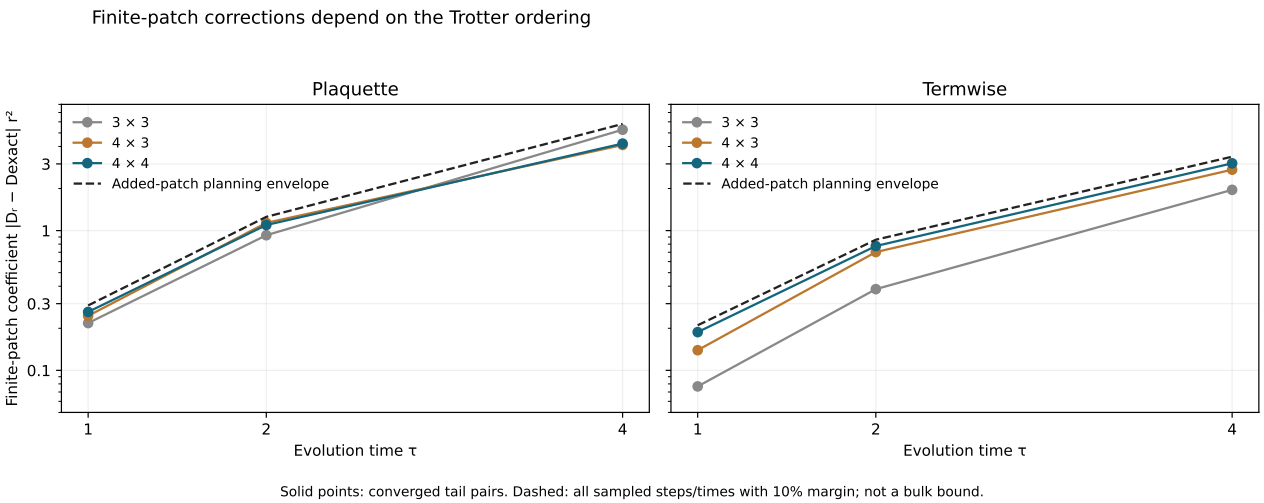


Figure 26: Ordering-specific Trotter coefficients through $\tau = 4$ on three finite patches. Solid points use converged tail pairs; the dashed envelope also covers measured coarser-step coefficients and earlier sampled times, with a 10% margin. The larger patches have only three time nodes; unsampled time peaks and larger-lattice corrections remain possible.

10 The fixed-device question

The same physical qubits can buy more simultaneous samples, stronger logical protection, or faster delivery of non-Clifford operations. We want to determine which use gives the largest accessible dynamics problem, and which combinations can improve on the three reference approaches. Figure 28 shows the resource roles. The comparison includes physical probabilistic error cancellation (PEC) [20], the original and small-angle STAR mechanisms [12, 4], and surface-code computation with explicit magic-state supply [19, 18]. Campbell's Hubbard construction provides a concrete algorithmic baseline [13]; recent partial-FT resource studies [7] and a performed fermionic-dynamics experiment [8] provide comparison points with distinct hardware assumptions.

We treat these as specified implementations and estimators. A poor physical PEC curve is not an impossibility theorem for NISQ, and an encoded curve based on reference patch timing is not a validated physical floorplan. The purpose of the analytical laws is to expose the quantities whose measurement or optimization can change the architecture ranking. CUDA-Q Logical supplies explicit circuit and synthesis artifacts [21]; physical encoded timing is kept separate when its implementation or verification is incomplete.

The similarity of the encoded data patches is physical: both approaches protect their Clifford operations. The main distinction is how they supply non-Clifford rotations, including the sequence of operations in time (Figure 31). STAR teleports an angle-specific resource state and mitigates residual rotation errors. Our FTQC reference instead synthesizes the rotation from Clifford gates and high-fidelity T states. In the original STAR preparation protocol, a surgery ancilla can also serve as the preparation patch when distance and physical error rate permit adequate acceptance [4, Sec. II A 2]. Preparation, any patch growth, and retries still occupy space and time. Dedicated injection lanes are therefore a scheduling choice in our cost model, not a defining STAR requirement. Conversely, FTQC supply may be compact or shared in time; neither approach has a universal ratio of data area to preparation area.

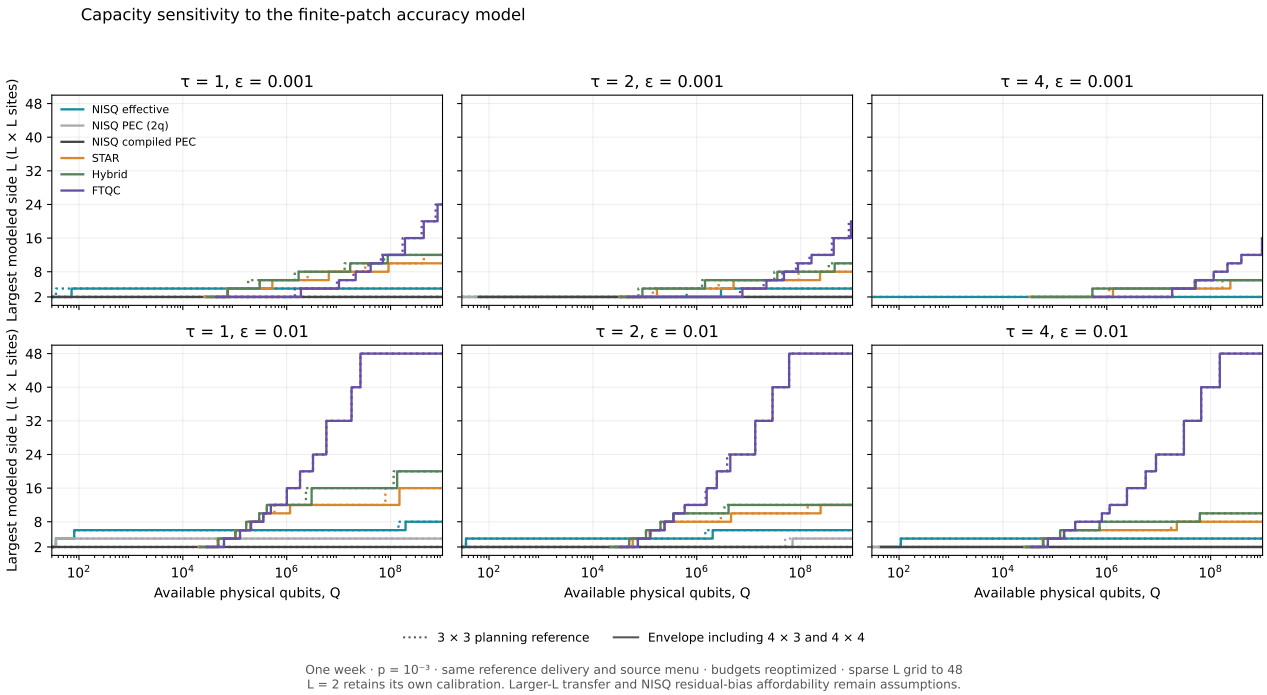


Figure 27: Largest modeled lattice side versus available qubits, changing only the finite-patch accuracy model. Dotted curves use the original 3×3 envelope; solid curves also include the 4×3 and 4×4 measurements. Delivery and source menus are identical, with error allocations reoptimized. Both policies retain direct $L = 2$ calibration. NISQ effective mitigation remains an assumed envelope; two-qubit-only PEC and the full compiled-noise PEC proxy are shown alongside it. All curves use a one-week cutoff, $p = 10^{-3}$, and the same sparse size grid through $L = 48$. A missing curve means that no sampled lattice fits that panel's qubit and time budgets.

11 Task, conventions, and precision

The problem has $N = L^2$ Hubbard sites, $2N$ spin modes, and open boundaries. The physical device has a separate area budget Q. In units where energies are measured in J, we simulate

$$H/J = -\sum_{\langle ij\rangle,\sigma} (c^\dagger_{i\sigma} c_{j\sigma} + \text{h.c.}) + u \sum_i (n_{i\uparrow} - \tfrac{1}{2})(n_{i\downarrow} - \tfrac{1}{2}), \qquad u = U/J = 4. \tag{31}$$

The symmetric onsite form differs from $U n_\uparrow n_\downarrow$ only by a known phase in a fixed-number sector. We prepare the half-filled Néel product state and measure the central doublon $D = \langle n_{i\uparrow} n_{i\downarrow} \rangle$ at $\tau = Jt/\hbar$. Doublon avoids the symmetry-pinned charge-density observable of this quench.

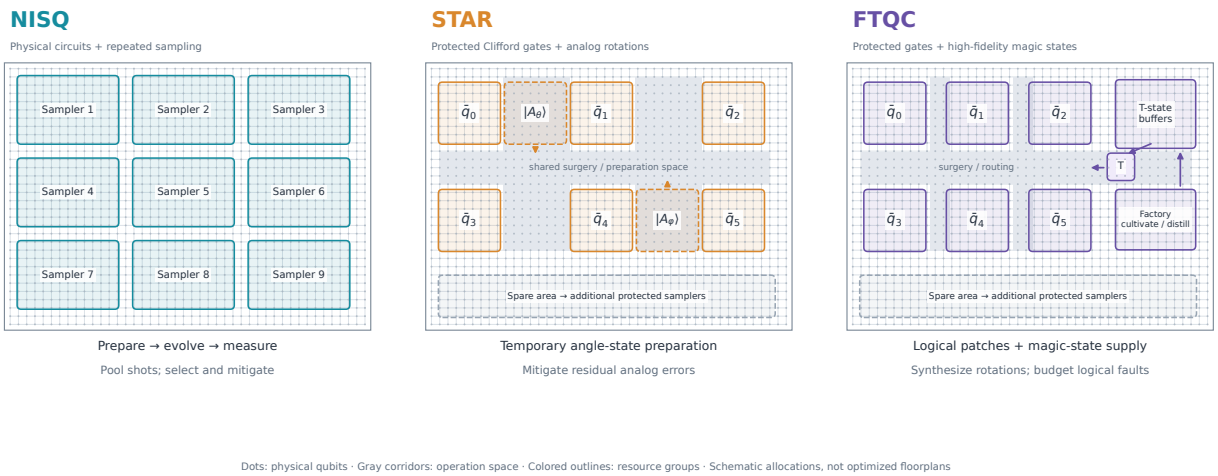


Figure 28: Three allocations of a common sea of physical qubits. NISQ uses many unprotected circuit replicas. STAR and FTQC share protected data patches and lattice surgery. Dashed orange patches show temporary angle-state preparation in STAR surgery ancillas, a possible allocation rather than a dedicated factory requirement [4]. FTQC supplies high-fidelity T states to a synthesized sequence of logical operations. Every dot denotes a physical qubit; each encoded data patch contains many such qubits. The drawings explain resource roles and are not numerical floorplans or equal-capacity implementations of a particular Hubbard lattice.

Quantity	Reference value or convention
Connectivity	Square array, nearest neighbors
Gate / measurement / reset	10 / 100 / 100 ns
Physical two-qubit / one-qubit error	$p = 10^{-3}$ / $p/10$
Idle channel	Depolarizing error p per 100 ns
Encoded cycle / classical feedforward	240 / 200 ns, reference assumptions
Runtime cutoff	$B = 604800$ s, one week
Main precision	1% relative at provisional $D_* = 0.1$: $\epsilon = 0.001$
Statistical confidence	$1 - \delta = 0.95$
Rotation convention	$U_P(\theta) = e^{-i\theta P}$; CUDA-Q angle $\phi = 2\theta$

The native physical compilation assumes arbitrary-angle R_{ZZ} and one-qubit pulses at the stated gate duration. The noise channels and encoded cycle are explicit model choices, not a characterization of a fabricated device. In particular, the surface-code suppression fit has not been calibrated to the same physical idle and syndrome schedule.

The target is $\epsilon = \eta|D|$ with $\eta = 0.01$. Resource curves use the declared signal D_*, rather than silently treating it as calculated dynamics. Campbell's energy precision proportional to N corresponds to fixed absolute precision in energy density [13]. Our local D is already intensive: fixed absolute error in D does not itself become stricter with size. Relative precision is useful when the signal magnitude is the relevant scale. Strict relative precision at zero signal needs a separate prescription; a production estimator with unknown nonzero D would require a pilot or sequential confidence procedure. We retain $\epsilon = 0.01$ absolute as a separately labeled sensitivity.

We allocate fractions $(0.5, 0.25, 0.1, 0.075, 0.075)$ of ϵ to statistical, algorithmic, logical, synthesis,

and magic-state errors. For a spatial patch Ω and product formula with r steps,

$$|\widehat{D} - D_{\text{target}}| \le \epsilon_{\text{stat}} + |D_{\Omega,r} - D_{\Omega}| + |D_{\Omega} - D_{\text{target}}| + \epsilon_{\text{logical}} + \epsilon_{\text{synth}} + \epsilon_{\text{magic}}. \tag{32}$$

The truncation scenario splits the existing 0.25ϵ algorithmic reserve equally between Trotter and spatial bias. It does not add a new allowance. Mitigation assumes exact channel knowledge; a residual calibration bias would need its own allocation within Eq. (32). No unverified postselection bias reduction is credited in a resource curve.

12 Sampling, replication, and maximum lattice size

For inverse channels $\mathcal{E}_j^{-1} = \sum_a c_{ja}\mathcal{P}_{ja}$, define $\gamma_j = \sum_a |c_{ja}|$, $\Gamma = \prod_j \gamma_j$, and $\chi = \log \Gamma^2$. Centering the doublon observable as $D - I/2$ puts the PEC sample in $[-\Gamma/2, \Gamma/2]$. Hoeffding's bound then gives

$$n \ge \frac{\Gamma^2}{2\epsilon_{\text{stat}}^2} \log \frac{2}{\delta}, \qquad N_0 = \left\lceil \frac{\log(2/\delta)}{2\epsilon_{\text{stat}}^2} \right\rceil. \tag{33}$$

At strict reference precision, $N_0 = 7377759$. This is a sufficient common sampling convention, not a claim that every estimator needs this many shots.

Let q_a and t_a be the physical area and time of one complete sampler, and let a be whole-circuit acceptance. With $m = \lfloor Q/q_a \rfloor$ replicas,

$$T_{\text{answer}} \simeq \frac{N_0 e^{\chi} t_a}{am}, \qquad Q_{\min}(L) = q_a \max\left(1, \left\lceil \frac{N_0 e^{\chi} t_a}{aB} \right\rceil\right). \tag{34}$$

The code excludes a sampler whose single attempt exceeds B. Resource-state rejection is charged in delivery time, not counted again as whole-circuit rejection. Integer replicas are retained where they matter. Packing is an area-budget reference; a global square placement, calibration overhead, and control bandwidth are not yet modeled in the encoded curves. For PEC, $Q_{\min}$ is the minimum replica area that meets this sufficient sampling bound for the chosen implementation. It is not a lower bound on the qubits needed by another estimator, or even a tight cost for this one.

The principal output is now

$$L_{\max}^{\text{tested}}(Q) = \max\{L \in \mathcal{L}_{\text{tested}} : \min_s Q_{\min}(L, s) \le Q\}, \tag{35}$$

where s labels allowed distances, supply configurations, and lane counts. This is a discrete upper envelope, not a smooth inverse fit. No feasible tested lattice is left blank, and reaching the largest tested size is marked as a survey boundary. We do not interpolate untested sizes or call that boundary a physical saturation. The companion output $\tau_{\max}(Q, L)$ follows by sweeping evolution time under the same contract; it has not yet been accuracy-qualified.

For a fixed circuit and estimator, Eq. (34) gives the exposure limit

$$\chi \lesssim \log \frac{amB}{N_0 t_a}. \tag{36}$$

Increasing hardware or runtime therefore buys logarithmic exposure, even though it can move a discrete lattice-size threshold appreciably. This is the mitigation wall of the specified strategy, not a universal NISQ law.

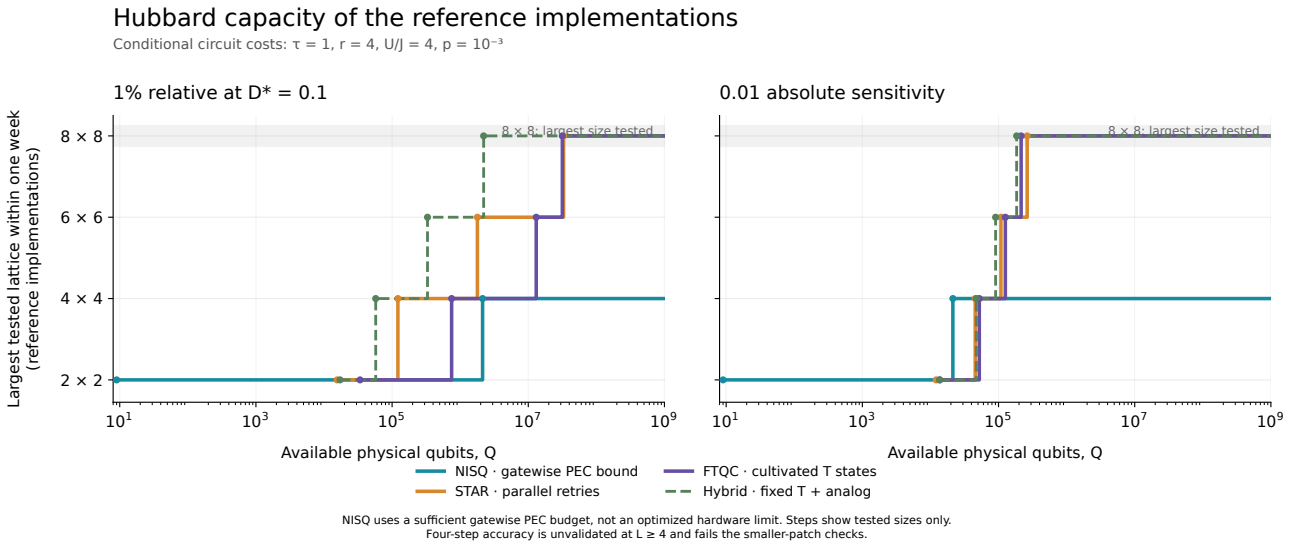


Figure 29: Largest tested Hubbard lattice within one week versus available physical qubits. Left: the accepted relative-precision reference. Right: the loose absolute-precision sensitivity. NISQ uses a sufficient gatewise-PEC sampling bound on the compiled observable cone at nominal idle noise; it does not represent optimized NISQ capacity. STAR chooses among one, two, four, and eight reference injection lanes; FTQC and hybrid use the packed cultivation supply scenario with zero additional handoff error. These are conditional $r = 4$ costs, not accuracy-qualified dynamics capacities. That step count fails the small square-patch checks, and $L \geq 4$ accuracy is uncalibrated. The NISQ compiler includes exact observable reductions that the encoded workloads have not yet received. Only $L = 2, 4, 6, 8$ is tested; the upper band marks that limit.

13 The common algorithm and its angle inventory

13.1 Termwise second order

In a fixed-number sector, the single-Z terms of the onsite interaction are a known phase. With $E = 2L(L-1)$ spatial bonds, the remaining Jordan–Wigner Hamiltonian has

$$K = 4E + N = 9N - 8L, \qquad \lambda = \sum_{\ell} |h_\ell| = 2E + \frac{|u|N}{4} = 5N - 4L. \tag{37}$$

There are four hopping Pauli strings per spatial bond, each with coefficient magnitude $1/2$, and one onsite ZZ per site, with magnitude $|u|/4$. Our convention is $U_P(\theta) = e^{-i\theta P}$; thus $R_z(\phi)$ has $\phi = 2\theta$, and a T gate has $|\theta| = \pi/8$ up to a global phase.

For a symmetric product over the K terms, choose onsite terms at both ends of the ordering and merge adjacent identical rotations at the center and between steps. Before further commuting or Clifford simplifications,

$$R_{\text{term}} = r(2K-2) + 1, \qquad \Theta_{\text{term}} = \tau\lambda. \tag{38}$$

The count follows from $r(2K-1)$ rotations after center merging and $r-1$ additional boundary merges. Every coefficient has total positive duration τ, proving the angle identity. Signed higher-order stages instead multiply the angle sum by their total absolute stage weight. The histogram, not merely Θ, is retained for channel evaluation.

13.2 Campbell plaquette second order with open boundaries

Campbell partitions hopping into disjoint pink and gold tiles and applies

$$S_2(\Delta) = e^{-i\Delta H_I/2} e^{-i\Delta H_p/2} e^{-i\Delta H_g} e^{-i\Delta H_p/2} e^{-i\Delta H_I/2}, \qquad \Delta = \tau/r. \tag{39}$$

We adapt the periodic construction to open boundaries. The pink partition has $P = L^2/4$ complete plaquettes. The shifted gold partition has $G = (L/2-1)^2$ complete plaquettes and $B = 2(L-2)$ isolated boundary bonds. Each bond belongs to exactly one partition, since $4(P+G)+B = E$.

A single-spin four-site hopping cycle has eigenvalues $\{2, 0, 0, -2\}$. Campbell's circuit implements its evolution with two variable-angle Pauli rotations and eight fixed T gates [13, Appendix E.1]. An isolated bond needs two commuting hopping Pauli rotations per spin. For $L \geq 4$, including both spin species and exact onsite boundary merging, this gives

$$R_{\text{plaq}} = N(r+1) + 4r(2P+G+B), \tag{40}$$

$$T_{\text{fixed}} = 16r(2P+G), \tag{41}$$

$$\Theta_{\text{dyn}} = \tau \left[\frac{|u|N}{4} + 4(P+G) + 2B \right]. \tag{42}$$

For $L = 2$ the gold partition is empty: the two pink half steps merge, so $2P$ becomes P in the count formulas. The angle formula is unchanged. At $L = 4$, $\tau = 1$, $r = 4$, these formulas give $(R_{\text{plaq}}, T_{\text{fixed}}, \Theta_{\text{dyn}}) = (288, 576, 44)$.

The compiled total angle includes $T_{\text{fixed}}\pi/8$. Consequently, the invariance of $\tau\lambda$ under refinement in Eq. (38) does not apply to all non-Clifford operations in Eq. (39). The fixed basis changes are an algorithm-dependent STAR exposure even when the variable rotations have no injection floor. Distilling those fixed angles is an immediate hybrid candidate. Hamming-weight phasing, cancellations between neighboring basis changes, and angle extraction remain further compiler optimizations. Campbell neglects fermionic-swap Clifford costs in his non-Clifford metric; our native nearest-neighbor compilation explicitly restores this motion.

14 Physical computation: gates, locality, and idle exposure

For a k-qubit depolarizing location with total nonidentity-Pauli probability p_k, set $s_k = 1 - 4^k p_k/(4^k - 1)$. The inverse has identity coefficient $1 + p_k/s_k$ and $4^k - 1$ coefficients $-p_k/[(4^k-1)s_k]$, giving

$$\gamma_k = 1 + 2p_k/s_k, \qquad \chi_{\text{phys}} = \sum_{\ell \in \text{cone}} 2\log\gamma_{k_\ell} \simeq 4pG_2 + 4(p/10)G_1 + 4p_{\text{idle}}G_{\text{idle}} + \chi_{\text{SPAM}}. \tag{43}$$

Preparation and readout bit flips use $\gamma = 1/(1-2p)$. The continuous-time idle convention composes as

$$p_{\text{idle}}(\Delta t) = \frac{3}{4}\left[1 - (1-4p/3)^{\Delta t/(100\,\text{ns})}\right]. \tag{44}$$

Only channels in the compiled measurement's backward dependency cone enter the local PEC product. CPTP channels outside that cone cannot affect its expectation. This sufficient cone can still exceed the observable support needed at a specified tolerance.

14.1 Fermionic blocks and parallel motion

Two-mode hopping, density evolution, and fermionic Hadamards are extracted from the plaquette workload while preserving its noncommuting stage order. A number-conserving Gaussian gate has odd-sector matrix $G \in U(2)$, vacuum amplitude one, and double-occupation amplitude $\det G$. An

Euler decomposition of G therefore compiles to local Z rotations and at most two native R_{ZZ} gates. In particular,

$$V(\theta) = e^{-i\theta(XX+YY)}, \qquad F_{\text{swap}} = (S \otimes S)V(\pi/4), \tag{45}$$

so a swap followed by hopping can fuse into the same pair of entanglers with a changed angle. Products of Gaussian gates on the same adjacent pair fuse through operations on disjoint wires. A commuting density rotation adds at most one entangler.

The compiler gathers independent plaquette groups into intervals of a live fermionic order using parallel odd-even transposition sorting. All swaps are adjacent physical fermionic swaps on a snake embedded in the square grid [22]. It then executes disjoint groups concurrently and tries six initial-order/local-route choices. A preparation permutation is free for the specified occupation product state, while the final wire map and Fock signs remain explicit. Using only snake edges leaves room for a better two-dimensional compiler. The trapped-ion experiment's all-to-all connectivity and free ordinary-SWAP relabeling are a different compilation assumption [8].

The first onsite stage is a scalar on the initial Fock state, and the last onsite stage commutes with the measured doublon. Removing them is exact for this task. A reverse traversal through even fermionic blocks then removes blocks on disjoint support. These reductions are available to every arm; the current encoded resource curves retain the original full logical workload.

$L = 4,\ \tau = 1,\ r = 4$	Entanglers	10 ns layers	Local χ
Greedy fused route with cone	3149	2187	35.297
Parallel route, full evolution	3204	1074	22.214
Parallel route with observable reductions	2648	964	20.017

The final row has $t_a = 9.84$ μs and exposure

$$\chi = 10.5927_{2q} + 4.1201_{1q} + 5.1681_{\text{idle}} + 0.1361_{\text{SPAM}}. \tag{46}$$

Its strict-precision threshold is $Q_{\min} \simeq 2.13 \times 10^6$, versus 12168 with zero idle on the same circuit. Idle has decreased substantially but remains material. The gain combines scheduling, placement, fusion, and observable algebra; zero idle is a sensitivity rather than a demonstrated alternate schedule.

14.2 What the million-qubit estimate means

The 4×4 circuit occupies 32 active qubits, packed into a 36-qubit square. At the strict precision, the reference assigns $\Gamma^2 = e^{20.017} \simeq 4.94 \times 10^8$ and a sufficient budget of $N_0\Gamma^2 \simeq 3.64 \times 10^{15}$ shots. At 9.84 μs per attempt, this requires 59239 copies to fit within one week, giving $Q_{\min} = 2132604$. The large area is entirely a sampling-throughput cost. For the same circuit, duration, and precision, removing idle noise gives $Q_{\min} = 12168$; retaining only two-qubit noise gives $Q_{\min} = 180$. These are noise ablations, not demonstrated improvements or alternative predictions for the declared device.

Experiments require an additional calibration of the NISQ model. Phasecraft and Quantinuum simulated a 7×4 lattice with 56 qubits, using training with fermionic linear optics (TFLO) and time-series smoothing [8]. The Willow study reached 6×6 with 72 fermionic-mode qubits, using TFLO, symmetry selection, and other application-specific processing [10]. Our gatewise PEC bound does not describe these estimators. Nor do those demonstrations establish our particular 0.001 absolute-error contract at every interacting data point. Their compilation is also a relevant benchmark: Willow's published 4×4 formula gives 1620 CZ gates at four steps, where our reference uses 2648 native R_{ZZ} gates. This comparison needs the respective gate implementations and schedules, not only a gate-count ratio. We therefore retain the current curve as a specified PEC reference; the resource frontier for experimentally informed physical estimators remains unestablished.

14.3 Exact circuit cones and physical truncation

For the original plaquette block ordering, the sufficient interior cones checked at $r = 1, 2, 4$ contain $(4r + 2)^2 = 36, 100, 324$ sites. The $r = 1$ circuits at $L = 8, 12, 16, 20$ all compile to 2001 native entanglers on 72 modes in that implementation. Ordered termwise sweeps can instead have global strict dependencies even at one step. Neither observation establishes the smallest Hamiltonian region needed at finite tolerance.

At fixed evolution time a local-observable truncation may eventually remove size dependence, whereas a time family such as $\tau \propto L$ probes growing physical reach. A truncation must receive a bias allowance in Eq. (32); it cannot be inferred solely from the strict circuit cone. Section 18 begins the finite-patch check and states what its outer boundary leaves unresolved.

15 Encoded protection and the logical-fault budget

Both encoded arms use the declared per-patch, per-cycle suppression reference

$$p_L(d) = A(p/p_{\mathrm{th}})^{(d+1)/2}, \qquad A = 0.1, \quad p_{\mathrm{th}} = 0.01. \tag{47}$$

The alternative $(A, p_{\mathrm{th}}) = (0.07, 0.0039)$ is motivated by the original STAR fit [12]. Its time convention and circuit noise must be reconciled before it becomes a calibration of this device. We use 240 ns per syndrome cycle and roughly $2d^2$ physical qubits per tile as reference accounting. Figure 30 distinguishes physical qubits, logical patches, and operation/supply space.

For V_L patch-cycles per attempt, $q_L \le V_L p_L(d)$ by a union bound. A $[0, 1]$ unmitigated observable has fault bias at most q_L. A uniform branch-wise bound for centered PEC gives the sufficient worst-case Γq_L. Thus the reference chooses the smallest admissible odd d with

$$V_L p_L(d) \le \epsilon_L/\Gamma, \qquad d \gtrsim \frac{2\log(AV_L\Gamma/\epsilon_L)}{\log(p_{\mathrm{th}}/p)} - 1. \tag{48}$$

The implementation solves the discrete self-consistency problem because duration and volume also depend on d. Faults are budgeted per attempt; the full sampling campaign is not required to contain zero logical faults. Commuting, branch-independent fault channels can justify removing Γ locally. Section 16.1 checks this condition for a declared logical injection instrument, and Appendix B explains why independence alone is insufficient.

16 STAR: angle exposure, floors, and delivery

For independent residual dephasing q_j after completed rotations,

$$\Gamma^2 = \prod_j (1 - 2q_j)^{-2}, \qquad \chi_{\mathrm{STAR}} = -2\sum_j \log(1 - 2q_j) \simeq 4\sum_j q_j. \tag{49}$$

Original STAR uses the post-RUS reference $q_0 = 4p/15$ [12]. The small-angle reference uses $q_j = \alpha p|\theta_j| + f$, with $\alpha = 1.5$ in our angle convention and a declared floor $f = 10^{-5}$ [4, 7]. The floor is a sensitivity motivated by injection and growth, not a measured error floor at this operating point. Fixed $\pi/8$ rotations retain q_0; we do not extrapolate the small-angle fit to them. Consequently,

$$\chi_{\mathrm{STAR}} \simeq 4\alpha p\Theta_{\mathrm{dyn}} + 4fR + \frac{16p}{15}T_{\mathrm{fixed}}. \tag{50}$$

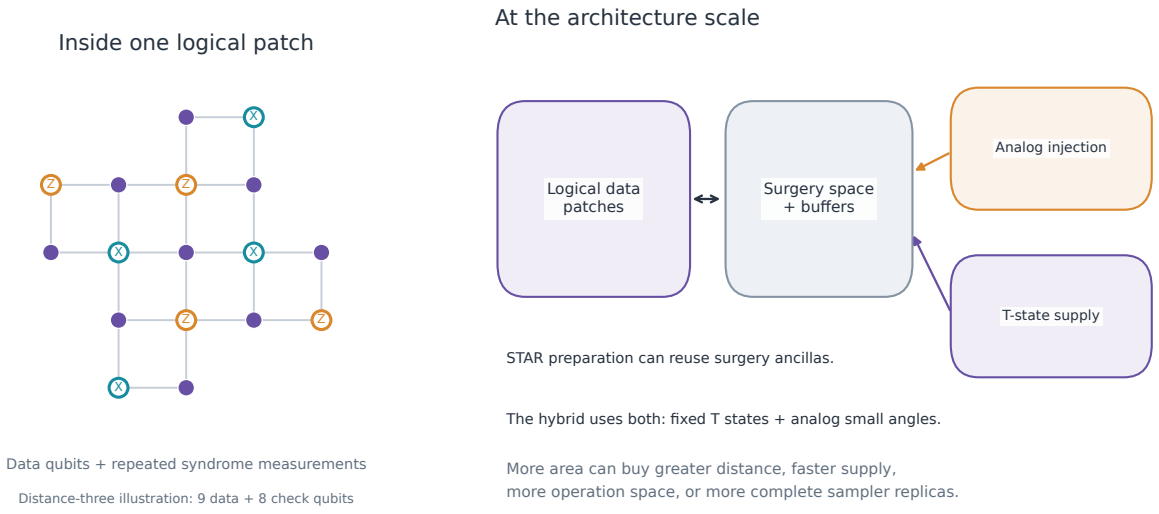


Figure 30: Physical and logical resource scales. Left: a distance-three rotated-code illustration with nine data and eight check qubits; links are nearest-neighbor data–check interactions. Right: logical data interacts through operation space with analog injection or T-state supply. STAR and FTQC use different supply paths; the hybrid uses both. The diagram shows functional roles: STAR preparation can occupy surgery ancillas rather than a separate supply region. This illustration is not a choice of distance or a calibrated surgery schedule for the resource curves.

The implementation evaluates the logarithmic product in Eq. (49); the physical channel models themselves remain approximations. Their three exposures have different responses to compilation and refinement. In particular, $fR \sim \alpha p \Theta_{\mathrm{dyn}}$ marks a floor/angle competition, while repeated basis changes can dominate separately.

The serial reference allocates $n_{\mathrm{data}} = \lceil 1.5(2N+6) \rceil$ tiles and six injection tiles. It charges 18 code clocks per rotation, with one clock equal to d syndrome cycles, plus startup, readout, and feedforward. This is a delivery model. A small-angle error law does not itself establish its acceptance, throughput, or footprint. The explicit retry refinement below adds both parallelism and its area cost.

16.1 Explicit retries, covariance, and parallelism

The logical injection instrument prepares $|A_\theta\rangle = e^{-i\theta Z}|+\rangle$, applies $\mathrm{CNOT}_{\mathrm{data}\to\mathrm{ancilla}}$, and measures the ancilla in Z. Its data Kraus operators are

$$K_0 = 2^{-1/2} e^{-i\theta Z}, \qquad K_1 = 2^{-1/2} e^{+i\theta Z}. \tag{51}$$

After outcome one, an injection of angle 2θ corrects a successful next branch: $K_0(2\theta)K_1(\theta) = K_0(\theta)/\sqrt{2}$. Repeating this gives the standard angle-doubling protocol underlying the STAR reference [12, 4]. CUDA-Q P0 now contains explicit preparation, CNOT, measurement, conditional retries, and a returned exhaustion flag for caps $1, 2, 4, 8, 24$. This compilation does not supply a surface-code injection or patch-growth implementation.

For a Pauli fault on either wire immediately before or after CNOT, each conditional K remains diagonal or antidiagonal with equal-magnitude entries. Hence

$$K^\dagger K = I/2, \qquad KZ = \pm ZK. \tag{52}$$

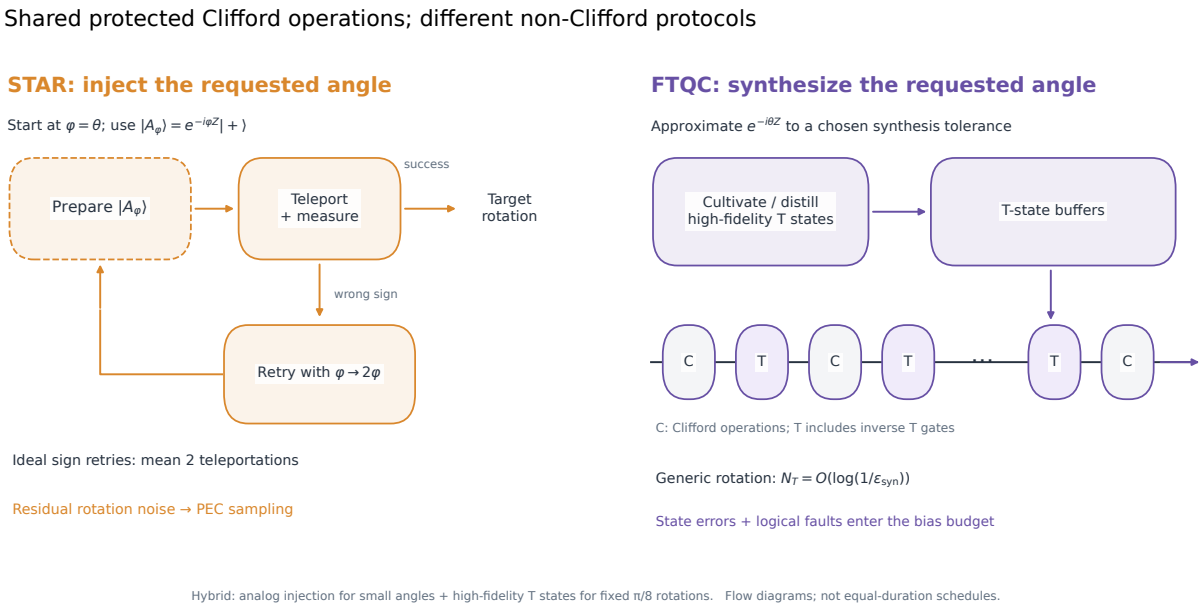


Figure 31: Two implementations of the same target rotation, in the convention $e^{-i\theta Z}$. STAR prepares an angle-specific resource state and uses teleportation with angle-doubling corrections [12, 4]. Ideal sign outcomes require two teleportations on average; this does not include rejected preparation attempts or their cost. FTQC consumes high-fidelity T states in a Clifford+T approximation, whose generic T count grows logarithmically with inverse synthesis tolerance [23]. The hybrid chooses between these paths by operation class. These are protocol flows, not schedules with equal durations or calibrated factory footprints.

The first identity makes the outcome probability independent of the data state; the second makes the conditional map commute with the channel $\mathcal{Z}(\rho) = Z\rho Z$. It therefore commutes with a Z-dephasing inverse, which is a linear combination of $\mathcal{I}$ and $\mathcal{Z}$. Mixtures of such faults and compositions over retries retain this local covariance under the stated branch-independent Pauli model. All sixteen two-qubit Paulis at both locations are checked independently.

This verifies the needed local commutation condition for that declared instrument. It does not establish branch-independent residual noise for physical state preparation, patch growth, measurement-dependent selection, or a mismatched channel inversion. The worst-case Γq_L bias bound therefore remains the default; q_L is a separately labeled covariance-based reference scenario. The physical STAR schedule still has to justify using it.

For b simultaneous ideal injections with geometric success probability $1/2$ and cap K, the mean batch retry time in units of one attempt is

$$\mathbb{E}[M_b] = \sum_{k=0}^{K-1} \Pr(M_b > k) = \sum_{k=0}^{K-1} \left[1 - (1 - 2^{-k})^b\right]. \tag{53}$$

It approaches 2 for one lane, but is approximately 3.50 for four lanes. Simply dividing the serial duration by four misses the slowest retry. The implemented policy has a barrier between batches, reserves six injection tiles per lane, and assumes nine clocks per attempt, recovering the previous 18-clock serial mean. It forms disjoint batches using full JW supports, not merely disjoint fermionic endpoints. An asynchronous policy could improve this conservative batch schedule.

With R rotations the ideal no-exhaustion probability is $(1 - 2^{-K})^R$. We choose K from a small allocated tail allowance and charge the acceptance penalty. At $L = 4$ and strict precision, two lanes

minimize area at $Q = 121716$, versus 166698 for the earlier serial reference. Four and eight lanes reduce attempt time further but lose on area. At loose precision one lane still wins. These results constrain the particular logical mapping and resource model, not the best possible STAR architecture.

17 Full fault tolerance and the hybrid

For R synthesized rotations each with operator error ζ, telescoping unitaries bound the observable bias by $2R\zeta$. We set $\zeta = \epsilon_{\text{syn}}/(2R)$. The deterministic synthesis fit

$$T_R = \lceil 3.02 \log_2(1/\zeta) + 0.77 \rceil, \qquad N_T = T_{\text{fixed}} + RT_R, \tag{54}$$

has the expected logarithmic scaling [23]. At the compiled points, actual CUDA-Q GridSynth words replace this estimate. At $L = 4, \tau = 1, r = 4$, termwise Strang has 889 variable rotations, no fixed T gates, and 61178 synthesized T gates. Plaquettes have 288 variable rotations and 576 fixed T gates, for 19584 synthesized T gates: 66 per variable rotation. Independent matrix reconstruction checks the precision of every distinct emitted word. Whole-program synthesis at $L = 2, 4$ agrees with the class-weighted counts; $L = 6, 8$ use those counts before inter-gate optimization.

17.1 Conventional factory reference

Injected-state error p_{in} is separate from gate error p. The current conventional reference uses $46p/15$, with alternatives retained as sensitivities; this coefficient is not calibrated to our asymmetric $p_{1q} = p/10$ noise model. For a 15-to-1 stage [19],

$$a_1 = 1 - 15p_{\text{in}}, \quad q_1 = 35p_{\text{in}}^3, \quad q_2 = 35q_1^3. \tag{55}$$

One stage reserves twelve tiles, including its output, and takes $11/a_1$ mean code clocks. The deliberately serial two-stage reference reserves two eleven-tile blocks and fifteen storage tiles, so

$$n_F = 37, \qquad c_F = \frac{15(11/a_1) + 11}{1 - 15q_1}. \tag{56}$$

Both stages use the selected code distance, and protected factory faults enter V_L. Correlations, transport, and rejection-tail latency are not resolved by this leading-order mean-service model.

For F factory copies, the FT timing reference is

$$C_{\text{FT}} = 2 + \max(9N_T + 2R, \lceil N_T/F \rceil c_F), \tag{57}$$

$$t_a = C_{\text{FT}} dt_c + N_T t_{\text{ff}} + t_{\text{reset}} + t_{\text{meas}}, \tag{58}$$

$$q_a = 2d^2(n_{\text{data}} + Fn_F). \tag{59}$$

Nine clocks per T is the compact-block consumption reference [19]; the $2R$ term is an extra Clifford-work placeholder. We optimize F over $\{1, 2, 4, 8, 16\}$ together with level and distance, subject to $N_T q_{1,2} \le \epsilon_{\text{magic}}$. Supply and consumption must be varied together: speeding consumption alone does not help a supply-limited circuit.

17.2 Why the hybrid is a separate candidate

The hybrid supplies the fixed T rotations as high-fidelity states while retaining analog small rotations. It removes the last term of Eq. (50), but pays factory area, supply time, and a magic-state budget

amplified by the retained analog PEC reference. At fixed ideal shot budget, a proposed additional protection step improves the wall-clock cost when

$$\Delta \log t_a + \Delta\chi - \Delta \log a - \Delta \log m < 0. \tag{60}$$

This provides a marginal criterion for operation-dependent protection, observable-aware allocation, and mixed rotation delivery. It must include the lost replicas when protection consumes area. The earlier hybrid gain at $L = 4$ reversed under conventional two-level injection supply; that was an amortization effect of the serial reference, not a general disadvantage of the hybrid. Cultivation changes the comparison again, for both FT and hybrid, under the supply assumptions that follow.

17.3 Cultivation for both FT and hybrid supply

Gidney, Shutty, and Jones report cultivated-state error approximately 2×10^{-9} at $p = 10^{-3}$ and, in their Fig. 15, expected volume 6.6×10^4 qubit-rounds for the distance-five protocol with early rejection and sharing [18]. The output is in a grafted matchable code after escape to nominal distance 15; its behavior is not identical to an ordinary distance-15 surface-code patch. Holding, transfer, growth, and consumption require additional accounting.

Our declared packed-supply scenario rounds the expected volume to $V = 10^5$ qubit-cycles, takes footprint $q_f = 1024$, successful latency $\ell_f = 50$ cycles, and charges a $2d$-cycle handoff. These latter choices are supply assumptions, not a compiled factory. The comparison also uses an unshared fixed-slot volume $V = 1024 \times 50/0.01 = 5.12 \times 10^6$. Mapping the paper's uniform physical noise and round definitions to this device's $p_{1q} = p/10$ and 240 ns cycle remains conditional.

For handoff error q_h, define

$$q_{\text{ready}} = 2 \times 10^{-9} + q_h, \qquad s_{\text{raw}} = \max(V/q_f, \ell_f) + 2d. \tag{61}$$

We sweep $q_h = 0, 10^{-9}, 10^{-8}$. Direct consumption is permitted only when

$$N_T q_{\text{ready}} \Gamma \le \epsilon_{\text{magic}}, \qquad \epsilon_{\text{magic}} = 0.075\epsilon, \tag{62}$$

with $\Gamma = 1$ for FT and the retained conservative analog-mitigation amplification for hybrid. Otherwise one additional 15-to-1 stage is available, with the same reference coefficients used previously [19]:

$$q_{\text{out}} = 35 q_{\text{ready}}^3, \qquad s_{\text{out}} = \frac{15 s_{\text{raw}} + 11d}{1 - 15 q_{\text{ready}}}. \tag{63}$$

The area includes the source footprint, one handoff buffer per source, and twelve further tiles if this stage is used. Surface-memory faults of the data, buffers, and final distillation stage enter the logical budget; the source preparation error already appears in Eq. (62). Supply and consumption run concurrently, so duration contains their maximum, and source copies and code distance are optimized under the week cutoff.

At strict precision and zero additional handoff error, direct unamplified consumption supports at most 37500 states. The compiled $L = 4$ FT count, 19584, fits; the $L = 6$ circuit requires the extra stage. Taking $q_h = 10^{-8}$ forces that stage already at $L = 4$. Cultivation therefore does not remove a distillation level independently of the task budget. In particular, the roughly 10^5 cutoff discussed around TopQAD Fig. 16 counts small-angle rotations at about 100 T states per rotation, not 10^5 T states [7].

$L = 4$, $r = 4$ reference	Q, $\epsilon = 10^{-3}$	Q, $\epsilon = 10^{-2}$
Serial STAR	166698	45486
FT, conventional serial factory	13791250	216890
FT, packed cultivation, $q_h = 0$	748656	52180
Hybrid, conventional serial factory	669714	54150
Hybrid, packed cultivation, $q_h = 0$	57472	47232

The conventional reference uses injected-state error $46p/15$. All rows remain conditional on $r = 4$. The loose-precision FT rows reuse the stricter synthesized words; resynthesis could improve them further. Cultivation changes the supply comparison appreciably, but neither these numbers nor their ranking should be promoted into a result for accurate dynamics.

18 Accuracy, truncation, and noisy observables

The calibration constructs H directly in fixed $(N_\uparrow, N_\downarrow)$ sectors. Sparse matrix exponentiation evaluates $e^{-i\tau H/J}$ without using the compiled gate decomposition. A separate evolution applies

$$U_r(\tau) = \left[e^{-i\Delta H_U/2} e^{-i\Delta H_{\text{pink}}/2} e^{-i\Delta H_{\text{gold}}} e^{-i\Delta H_{\text{pink}}/2} e^{-i\Delta H_U/2}\right]^r, \qquad \Delta = \tau/r, \tag{64}$$

where energies are in units of J. The largest patch is 3×3, with sector dimension $\binom{9}{4}\binom{9}{5} = 15876$. An independent eight-qubit dense Hamiltonian verifies the exact-sector observable; the original compiled plaquette circuit independently verifies the product formula on 2×2. The saved evidence contains 112 Trotter comparisons and 164 exact-curve points over four patches.

Patch, $\tau = 1$	Exact D	Error $r = 4$	Error $r = 16$	Error $r = 64$
2×2	0.1179790	0.0136397	0.00087075	0.00005447
3×3	0.1050849	0.0143939	0.00085350	0.00005317

For these square patches, $r = 64$ is the first tested power of two passing the strict split budget; $r = 16$ passes the loose split budget. These are not minimal-step claims. The asymptotic local error is consistent with r^{-2}, while coarse steps can be nonmonotonic. We also record a budget based on 1% of each actual patch signal, rather than silently replacing the provisional D_* in every resource curve.

The compiled $L = 2$ circuits at $r = 4, 16, 64$ contain respectively $164, 740, 3044$ native entanglers. The selected PEC estimator needs only one nine-qubit square copy at $r = 16$, but approximately 4.73×10^6 qubits at $r = 64$ to meet the week cutoff. Thus a common Trotter refinement can affect architecture costs very differently: it increases a physical noise exposure in an exponential, even when its algorithmic error reduction is shared. The corresponding $L = 4$ compilations demonstrate cost sensitivity only; their Trotter error has not been calibrated.

To start testing physical truncation, compare a five-site cross with its enclosing 3×3 square, keeping the same Néel pattern and central observable. Their exact differences are 4.75×10^{-4} at $\tau = 0.1$ and 7.93×10^{-2} at $\tau = 1$. Both exceed the strict truncation allowance. However,

$$|D_{\text{cross}} - D_{\text{target}}| \le |D_{\text{cross}} - D_{3\times3}| + |D_{3\times3} - D_{\text{target}}|, \tag{65}$$

and the second term is unknown. These finite-reference differences neither certify a bulk light cone nor bound the error of the larger patch. Larger patches, controlled local bounds, or both are still needed.

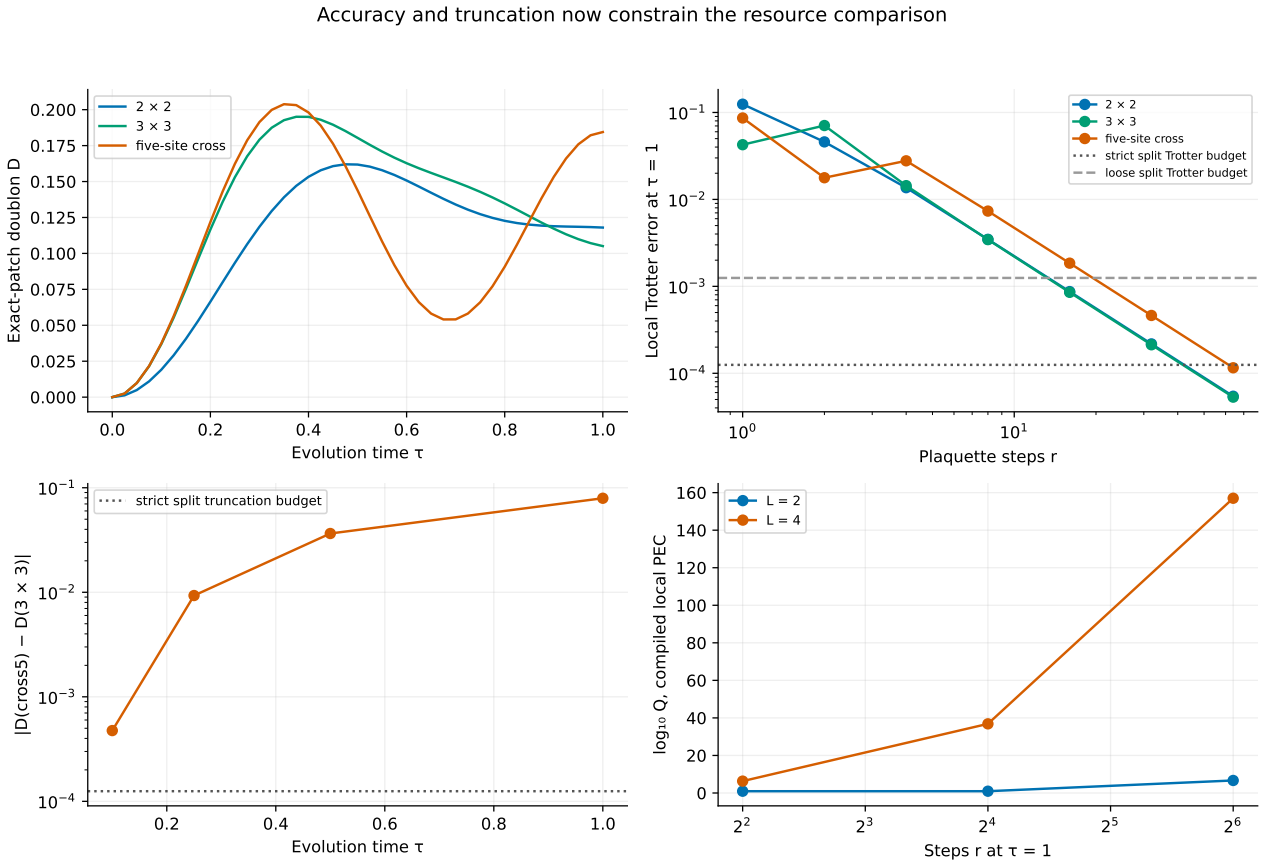


Figure 32: Exact patch dynamics, local product-formula errors, a nested spatial comparison, and the cost sensitivity to r. Each exact curve belongs to its own finite patch. The spatial difference is not an infinite-volume error certificate. The $L = 4$ cost sweep does not assume the small-patch error law holds at $L = 4$.

18.1 Postselection: measuring bias as well as acceptance

We propagate the full eight-qubit density matrix through the saved $L = 2$ native circuits. After each k-qubit primitive, the channel is

$$\mathcal{D}_{k,p}(\rho) = (1-p)\rho + \frac{p}{4^k - 1}\sum_{P \neq I} P\rho P. \tag{66}$$

Primitive idle channels, initial preparation flips, and final readout flips are also included. A direct Pauli-mixture implementation independently checks the optimized channel calculation. For selection event A, the measured quantity is $D_A = \mathbb{E}[D\mathbf{1}_A]/\Pr(A)$; a large or small acceptance alone says nothing sufficient about its bias.

At $r = 4$, the unselected noise bias relative to the ideal Trotter circuit is 0.020371. Selecting the two spin parities reduces it to 0.005808 with acceptance 0.81958. Selecting exact $N_\uparrow = N_\downarrow = 2$ reduces it to 0.004212 with acceptance 0.80943. Its bias relative to exact dynamics, including the independently measured Trotter error, is 0.01785. At $r = 64$, the number-selected noise bias rises to 0.1070. Postselection helps here, but by itself does not meet the strict precision target.

The parity acceptance agrees with the independently propagated Pauli-character formula. Trace

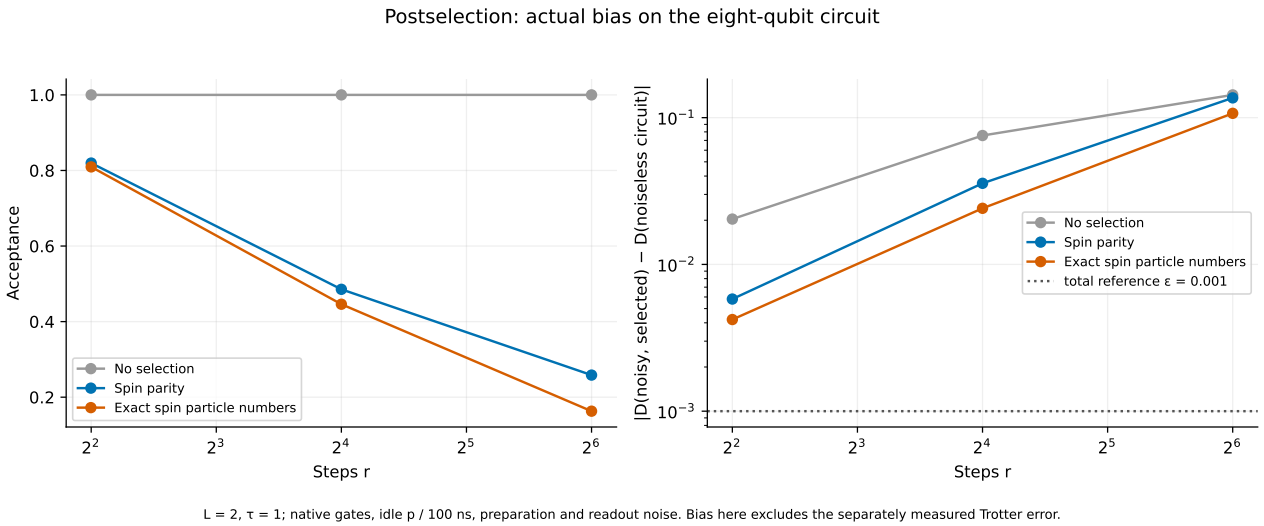


Figure 33: Exact acceptance and actual observable bias under the declared eight-qubit noise model. The right panel isolates noise bias from Trotter bias. No extrapolation from these curves to $L \geq 4$ is used.

errors are below 1.4×10^{-12} and the calculated density matrices remain positive. These results apply to the stated eight-qubit circuits and noise channels. They do not prove failure of selection at $L = 4$ or of combinations with other mitigation methods. The larger trapped-ion experiment [8] remains a separate circuit, geometry, and noise comparison.

19 What determines the scaling

The exposure variables give a more transferable comparison than a single crossover. With local full-lattice compilation, $R = O(Nr)$ and $\Theta = O(N\tau)$. Physical PEC and original STAR therefore have reference exponents $O(pNr)$, while the small-angle term has exponent $O(pN\tau)$. An injection floor and the plaquette fixed-angle basis changes restore $O(r)$ growth of the STAR exponent. Individual FT synthesis has $N_T = O(Nr \log(Nr/\epsilon))$ and logarithmic distance growth below threshold, but starts with a substantial protected footprint and supply overhead. These are laws of the specified models, subject to the locality and accuracy qualifications above.

The physical-error sweep in Figure 11 improves the modeled capacity of all three reference architectures as p falls. The relative ordering also depends on precision, injection floors, and source quality; holding these quantities fixed or assigning improvement laws gives different partial-FT windows. Figure 12 identifies an algorithmic change within FT that moves the strict-precision costs by approximately a factor of six. The compact-encoding scenarios and ordering diagnostic show why locality alone does not determine either the small-system gate count or the Trotter coefficient.

The earlier compiled reference in Figure 29 retains a separate, conditional circuit comparison. At $Q = 10^6$ and strict reference precision, the largest tested conditional circuits are 2×2 for NISQ, 4×4 for STAR and FTQC, and 6×6 for the hybrid. These numbers use different implemented optimizations, conditional encoded supply, and the same unqualified $r = 4$ product formula. They are not final architecture rankings. At loose precision the encoded thresholds cluster much more closely, making delivery rate, distance, and integer replicas consequential.

Small-patch calibration explains why a common algorithmic correction does not cancel from the resource comparison: increasing r reduces a shared Trotter error but enters each architecture's cost differently. Likewise, physical truncation could reduce all three arms if its bias is controlled. Direct

4×4 checks now settle the Trotter component for the selected headline workloads. The remaining scientific dependencies are transfer of accuracy to larger lattices, a controlled spatial truncation estimate, empirical NISQ mitigation bias, and physical injection/growth and handoff schedules with residual-channel validation. The same-Hamiltonian qubitization comparison in Section 4.1 does not reverse the tested ordering; its routed implementation and tighter state-dependent locality bounds remain unvalidated.

20 Conclusions

We compared error mitigation, STAR partial protection, and full surface-code fault tolerance for Fermi–Hubbard dynamics on one nearest-neighbor array, organizing the comparison by the largest time-growing workload that fits a qubit budget within one week. The common accounting model reproduced its earlier results and then optimized five choices that had been fixed. The resulting picture is the one the exposure variables predict. Physical-qubit computation saturates logarithmically in the device size; STAR and the hybrid saturate at a wall set by their accumulated rotation angle; and full fault tolerance pays a large fixed area and then keeps scaling. With delivery optimized, full fault tolerance separates at about 6×10^5 qubits at $p = 10^{-3}$ for both precisions studied.

Several limitations remain. Lane, port, and transfer charges are declared rather than compiled. The Trotter envelope is measured only through 4×4 patches and is transferred beyond $\tau = 4$. The calibrated surface-code law ($\Lambda \approx 7.4$ rather than 10 at $p = 10^{-3}$) is not yet propagated, the effective NISQ overhead is assumed, and the raw-RUS cancellation in our restricted instrument disagrees with the branch-averaged analysis of Ref. [4]. The most useful next calculations are a compiled schedule for parallel analog injection, propagation of the calibrated surface-code law, and interacting convergence checks on larger patches. Each could move a boundary; the parallel-injection schedule is the one that could reorder STAR and full fault tolerance.

A Matrix-free finite-lattice reference

Writing the state as a matrix indexed by up- and down-spin occupation masks, we apply

$$H\Psi = h_\uparrow \Psi + \Psi h_\downarrow^{\mathsf{T}} + V \odot \Psi. \tag{67}$$

Each hopping matrix is sparse within a single spin sector; no full-sector Hamiltonian is stored. Fermionic signs count occupied sites between the hopping endpoints in the snake order. For $U/J = 4$, the triangle bound is $\|H\| \le 2|E| + N = 64$ on 4×4. We evaluate

$$e^{-iH\tau}\psi_0 = J_0(a)\psi_0 + 2\sum_{k\ge 1}(-i)^k J_k(a) T_k(H/64)\psi_0, \qquad a = 64\tau, \tag{68}$$

using the three-term Chebyshev recurrence. The bound

$$|J_k(a)| \le \frac{(|a|/2)^k}{k!} \exp\!\left(\frac{a^2}{4(k+1)}\right) \tag{69}$$

follows by bounding the absolute Bessel series; successive right-hand sides have ratio at most $|a|/[2(k+1)]$. A geometric tail therefore selects a degree with state truncation error below 10^{-12}. For a $[0,1]$ observable, state error η gives observable error at most $2\eta + \eta^2$. This controls polynomial truncation, separately from floating-point error. The computed 4×4 norm error is about 10^{-13}.

Trotter hopping stages use exact two-state rotations. Plaquette stages use the fermionic Hadamard decomposition already employed by the compiler. Complete state vectors for the exact reference and

both orderings agree with the independent Python implementation on 2×2, 3×2, and 3×3 to better than 2×10^{-11}. Large-run outputs retain all-site doublons, norms, input and executable hashes, and atomic case records. The step convergence criterion is a 5% agreement between the two finest observable-error coefficients, not a proof of bulk convergence.

B Logical-fault commutation and its limitation

A branch-independent fault channel $\mathcal{F}$ outside a complete PEC cancellation block, or commuting through its noise channel $\mathcal{E}$, satisfies $\mathcal{E}^{-1}\mathcal{F}\mathcal{E} = \mathcal{F}$. For a stochastic fault of probability q_L, this gives bias at most q_L for a $[0, 1]$ observable. Pauli channels commute locally, making this a useful condition. It does not follow merely from linearity or branch independence.

For a counterexample, take Bloch matrices $E = \mathrm{diag}(s, s, 1)$ and $F = (1 - q)I + qH$, where H is Hadamard conjugation. On $|0\rangle$, measuring $(I + X)/2$ after $E^{-1}FE$ gives bias $q/(2s) > q$ when $s < 1/2$. This is a stochastic non-Pauli fault inside the cancellation sandwich. The logical STAR instrument of Section 16.1 instead verifies local Z covariance under its stated Pauli fault model. A physical implementation must justify that model across injection, growth, retries, and any selection before the more favorable budget becomes its default.

C QSVT approximation and selection budget

For $x = \alpha\tau$, the Jacobi–Anger expansion has coefficients $J_0(x)$ and $2(-i)^n J_n(x)$. Truncation after degree D has uniform operator-error bound $2\sum_{n>D} |J_n(x)|$. We sum the finite Bessel tail and bound the remainder using $|J_n(x)| \le (|x|/2)^n/n!$ and a geometric majorant. The even and odd parts approximate cosine and sine, each divided by two; the branch LCU divides by two again. The resulting block approximates $U/4$. We reserve half of its error allowance for polynomial truncation and half for classical phase realization. The latter is a required tolerance on phase generation, not a numerical validation of every large-degree phase list.

After $j = 0, 1, 2, 3$ oblivious amplification steps, the accepted singular-value polynomials are respectively

$$P_1(a) = a, \qquad P_3(a) = 3a - 4a^3, \tag{70}$$

$$P_5(a) = 5a - 20a^3 + 16a^5, \qquad P_7(a) = 7a - 56a^3 + 112a^5 - 64a^7. \tag{71}$$

Their ideal amplitudes are $a_j = \sin[(2j + 1)\arcsin(1/4)]$. For a block error δ, telescoping each matrix monomial gives $\eta_j \le C_j\delta$, where $C_j = \sum_n n|c_n|(1/4 + \delta)^{n-1}$ for $P_{2j+1}(a) = \sum_n c_n a^n$. The acceptance is at least $(a_j - \eta_j)^2$, and a conservative conditional observable-error bound is $4\eta_j/(a_j - \eta_j)$. Independent tests check both the amplification unitary and noncommuting matrix perturbations. Rotation synthesis receives a further operator-error allowance chosen with the same normalization bound. A stochastic circuit fault of probability q is charged at conditional bias $2q/P$ for acceptance lower bound P, rather than using the deterministic-circuit bound q. Shots include failed selection and a conservative fault-related acceptance reserve. Resource estimates retain the reference synthesis fit and source model, including cleanup when the selected budget requires it.

D A conservative centered truncation bound

In the interaction picture of the onsite terms, each even hopping operator remains supported on its two sites and keeps norm $2J$. Disjoint even fermionic operators commute, so the usual commutator recursion applies without interpreting Jordan–Wigner strings as physical propagation. Duhamel's

formula expresses the difference between full and truncated Heisenberg observables as an integral over omitted boundary bonds. Iterating the commutator recursion inside the retained patch produces a chain of overlapping bonds terminating at that boundary. A chain of length n contributes at most $\|D - I/2\|(4|\tau|)^n/n!$. There are at most $4 \cdot 7^{n-1}$ such chains before imposing the endpoint condition, and no boundary-reaching chain has $n < R + 1$. Summing gives Eq. (21). Time dependence in the interaction picture does not change the uniform norm or ordered-simplex bounds. For $x = 28|\tau|$ and $R + 2 > x$, the numerical implementation uses

$$\sum_{n \geq R+1} \frac{x^n}{n!} \leq \frac{x^{R+1}}{(R+1)!} \frac{1}{1 - x/(R+2)}. \tag{72}$$

The operator-norm bound also bounds the expectation difference for consistent initial reduced states. Values use double precision with a small positive roundoff margin; no interval-arithmetic certification is claimed.

E Compilation, checks, and reproducibility

The installed CPU CUDA-Q Logical environment compiles the common analog program, native physical circuits, and synthesized Clifford+T words. The physical routes carry mode permutations, explicit nearest-neighbor gates, ASAP schedules, counts, and hashes. The STAR retry program contains real logical preparation, measurement, and conditional angle doubling with an exhaustion flag. These P0 artifacts do not imply a physical STAR QEC schedule.

The installed minimal surface recipe lacks some needed gate gadgets. An explicit encoded macro adapter reaches physical IR, but the Hubbard backend timing is rejected. A reduced circuit with four fixed-angle rotations repeated four times, totaling 16 T states, triggers a nested schedule interval error. The 4-, 8-, and 32-T variants pass a loose bound, and separate two-resource template-call controls return their expected times. The root cause remains unresolved. None of these backend timings enters the resource curves; passing the loose controls is not a validation of physical encoded timing.

Tests independently check fermionic signs, product-formula identities, synthesis accuracy, exact-sector dynamics, primitive Pauli noise, STAR instrument algebra, and resource budgets. The capacity inversion additionally checks threshold inclusion, nonmonotone cost envelopes, and the untested region. These checks validate the specified calculations rather than the hardware assumptions.

The repository entry point is `README.md`. Source code is in `python/`, tests in `tests/`, assumptions in `configs/`, and this manuscript in `tex/main.tex` with a dedicated `tex/references.bib`. Current overview figures are in `figures/`; versioned scientific evidence stays in `results/`. Earlier complete notes are in `tex/archive/`. The original v3 manifest is preserved: historical source hashes are checked against their archived snapshots, while the relocated code replays the same saved numerical evidence. The presentation manifest records the new figures and their source datasets without relabeling older results as a new calculation.

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